

Econ 103: Project 1

Mason Lee, Alexander Prater, Fahad Malik, Kalani Leibow

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Packages

```
#install.packages("Lahman")  
library(Lahman)  
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.1.2
```

```
## Warning: package 'tibble' was built under R version 4.1.2
```

```
## Warning: package 'forcats' was built under R version 4.1.2
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.5
```

```
## v forcats    1.0.0      v stringr    1.5.1
```

```
## v ggplot2    3.5.1      v tibble     3.2.1
```

```
## v lubridate  1.9.3      v tidyr      1.3.1
```

```
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
```

```
library(car)
```

```
## Warning: package 'car' was built under R version 4.1.2
```

```
## Loading required package: carData
```

```
## Warning: package 'carData' was built under R version 4.1.2
```

```
##
```

```
## Attaching package: 'carData'
```

```
##
```

```
## The following object is masked from 'package:Lahman':
```

```
##
```

```
##      Salaries
```

```
##
##
## Attaching package: 'car'
##
## The following object is masked from 'package:dplyr':
##
##   recode
##
## The following object is masked from 'package:purrr':
##
##   some
```

Data Loading

```
data(Batting)
head(Batting)
```

```
##   playerID yearID stint teamID lgID  G AB R H X2B X3B HR RBI SB CS BB SO IBB
## 1 aardsda01  2004     1   SFN  NL 11  0 0 0  0  0  0  0  0  0  0  0  0
## 2 aardsda01  2006     1   CHN  NL 45  2 0 0  0  0  0  0  0  0  0  0  0
## 3 aardsda01  2007     1   CHA  AL 25  0 0 0  0  0  0  0  0  0  0  0  0
## 4 aardsda01  2008     1   BOS  AL 47  1 0 0  0  0  0  0  0  0  0  0  1  0
## 5 aardsda01  2009     1   SEA  AL 73  0 0 0  0  0  0  0  0  0  0  0  0  0
## 6 aardsda01  2010     1   SEA  AL 53  0 0 0  0  0  0  0  0  0  0  0  0  0
##   HBP SH SF GIDP
## 1   0  0  0   0
## 2   0  1  0   0
## 3   0  0  0   0
## 4   0  0  0   0
## 5   0  0  0   0
## 6   0  0  0   0
```

Data Manipulation

```
BattingData <- Batting

BattingData <- BattingData %>%
  filter(yearID >= 1985, AB >= 100, IBB > 0, HR > 0, SO > 0, RBI > 0) %>%
  mutate(BA = H / AB, Singles = (H - X2B - X3B - HR),
         SLG = ((H - X2B - X3B - HR) + 2 * X2B + 3 * X3B + 4 * HR) / AB,
         OBP = (H + BB + HBP) / (AB + BB + SF + HBP)) %>%
  mutate(OPS = SLG + OBP)

BattingDataSlim <- BattingData[c(1, 2, 5, 12, 13, 17, 18, 25)]

head(BattingDataSlim)

##   playerID yearID lgID HR RBI  SO IBB      SLG
## 1 abbotje01  1998   AL 12  41  28   1 0.4918033
```

```
## 2 abbotje01 2000 AL 3 29 38 1 0.3953488
## 3 abbotku01 1994 NL 9 33 98 1 0.3942029
## 4 abbotku01 1995 NL 17 60 110 4 0.4523810
## 5 abbotku01 1996 NL 8 33 99 1 0.4281250
## 6 abbotku01 1997 NL 6 30 68 3 0.4325397
```

Graphs of 4 Predictors (Question 2)

5 Number Summary

```
fivenum(BattingDataSlim$HR)
```

```
## [1] 1 5 11 19 73
```

```
fivenum(BattingDataSlim$RBI)
```

```
## [1] 3 29 49 73 165
```

```
fivenum(BattingDataSlim$SLG)
```

```
## [1] 0.1694915 0.3725490 0.4208038 0.4738806 0.8634454
```

```
fivenum(BattingDataSlim$S0)
```

```
## [1] 4 44 68 96 223
```

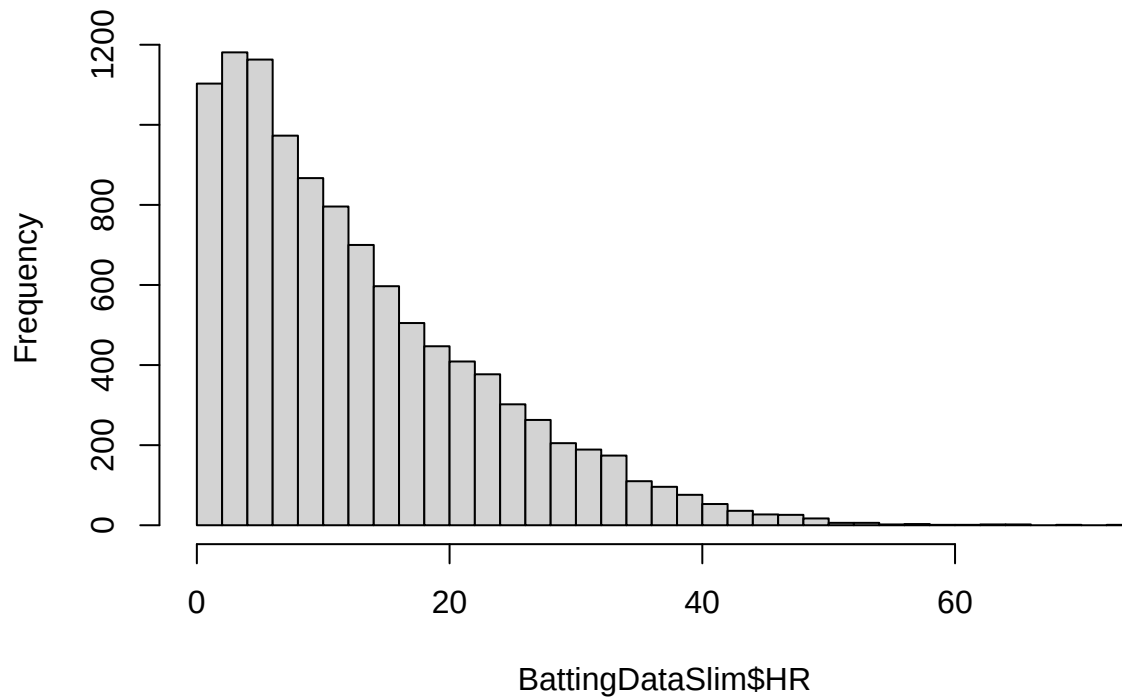
```
fivenum(BattingDataSlim$IBB)
```

```
## [1] 1 1 3 5 120
```

Histograms

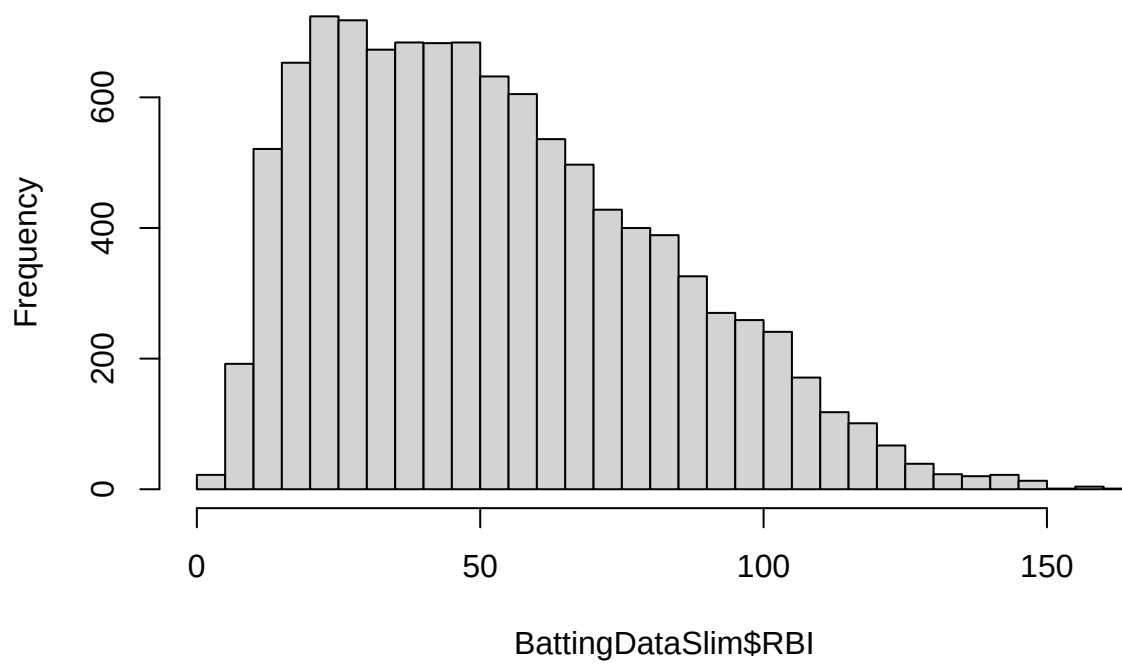
```
hist(BattingDataSlim$HR, breaks = "scott")
```

Histogram of BattingDataSlim\$HR



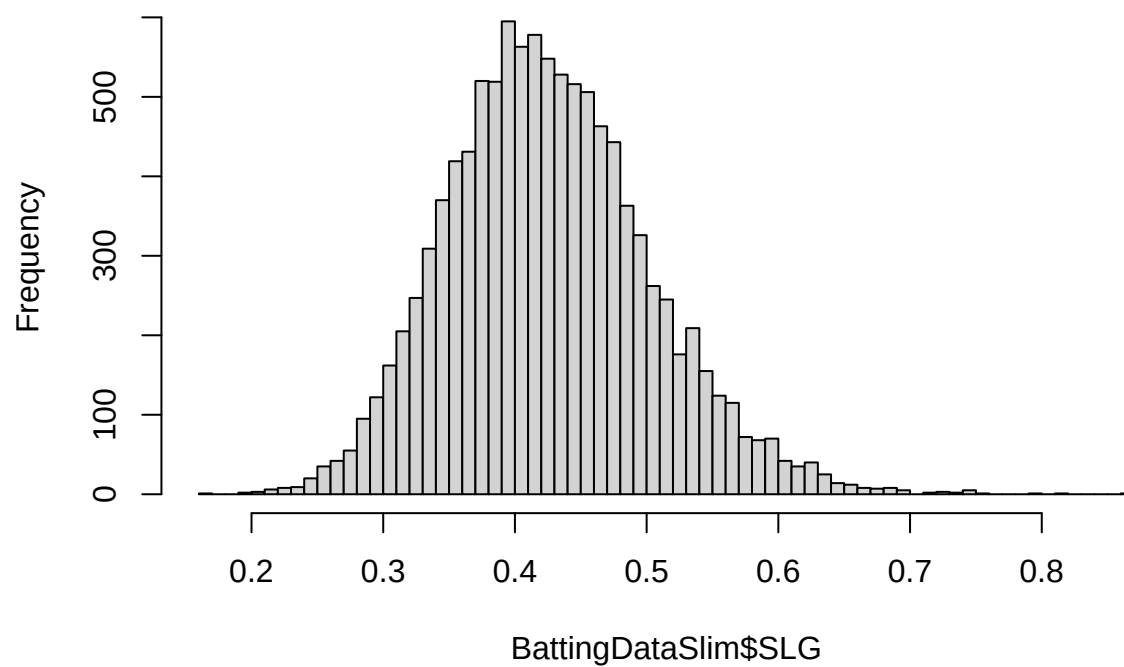
```
hist(BattingDataSlim$RBI, breaks = "scott")
```

Histogram of BattingDataSlim\$RBI



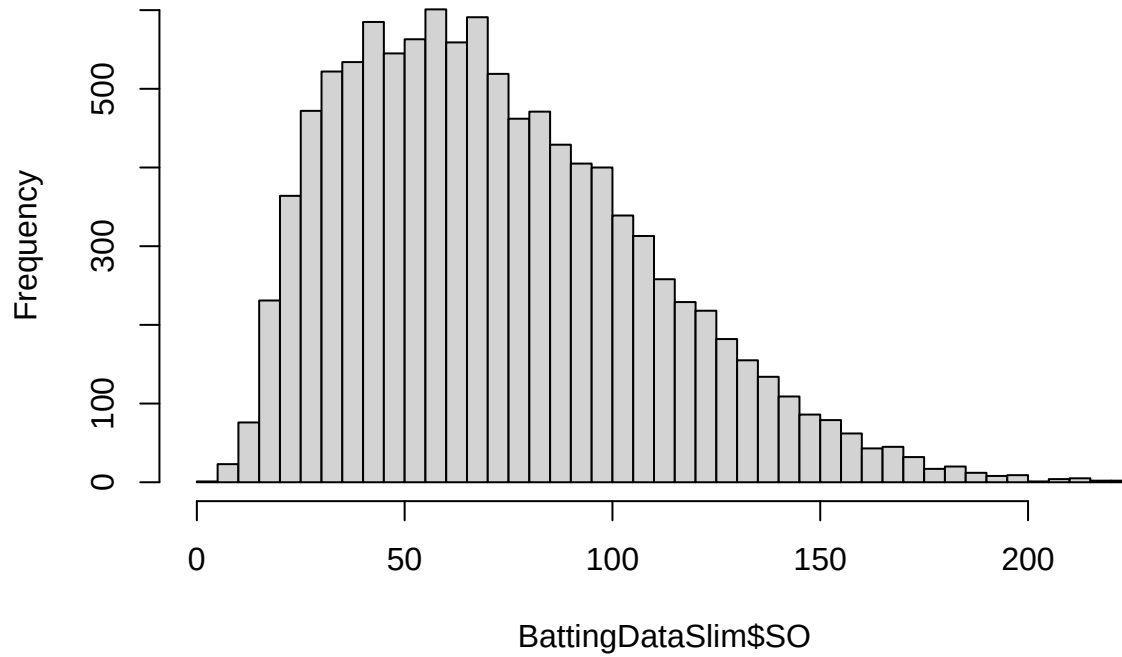
```
hist(BattingDataSlim$SLG, breaks = "scott")
```

Histogram of BattingDataSlim\$SLG



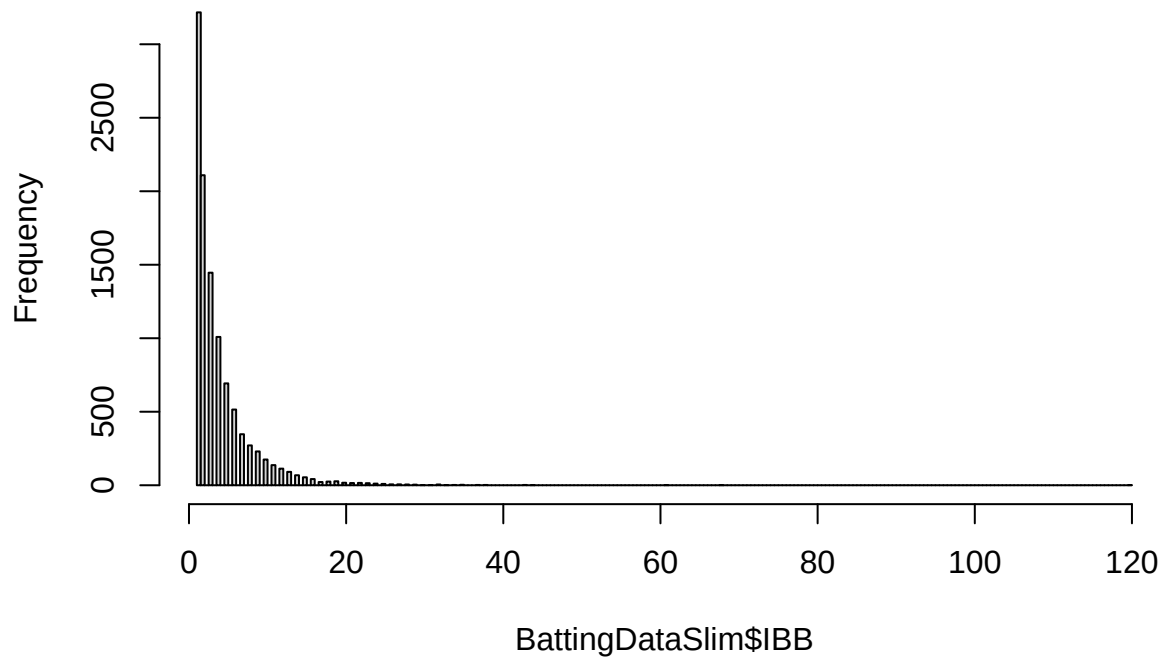
```
hist(BattingDataSlim$S0, breaks = "scott")
```

Histogram of BattingDataSlim\$SO



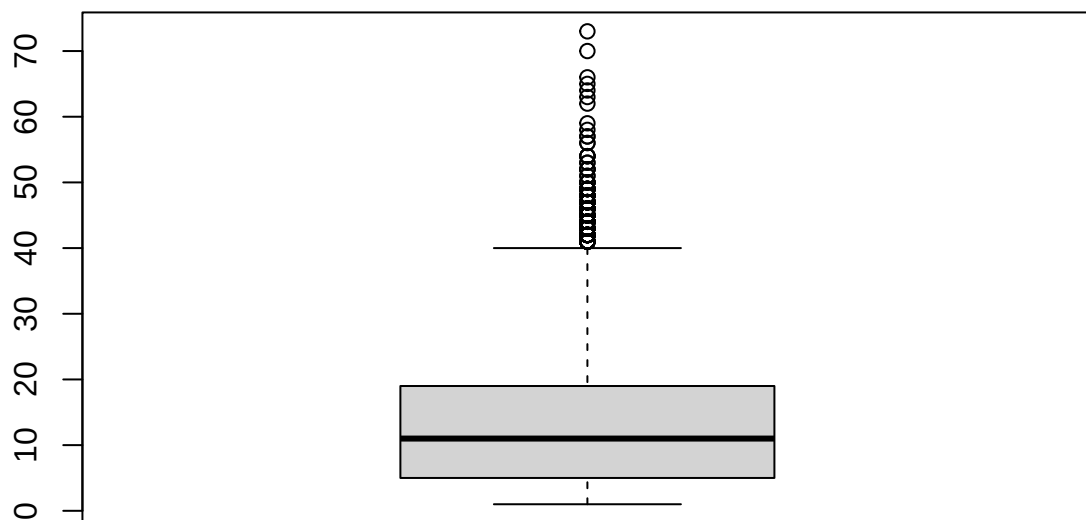
```
hist(BattingDataSlim$IBB, breaks = "scott")
```

Histogram of BattingDataSlim\$IBB

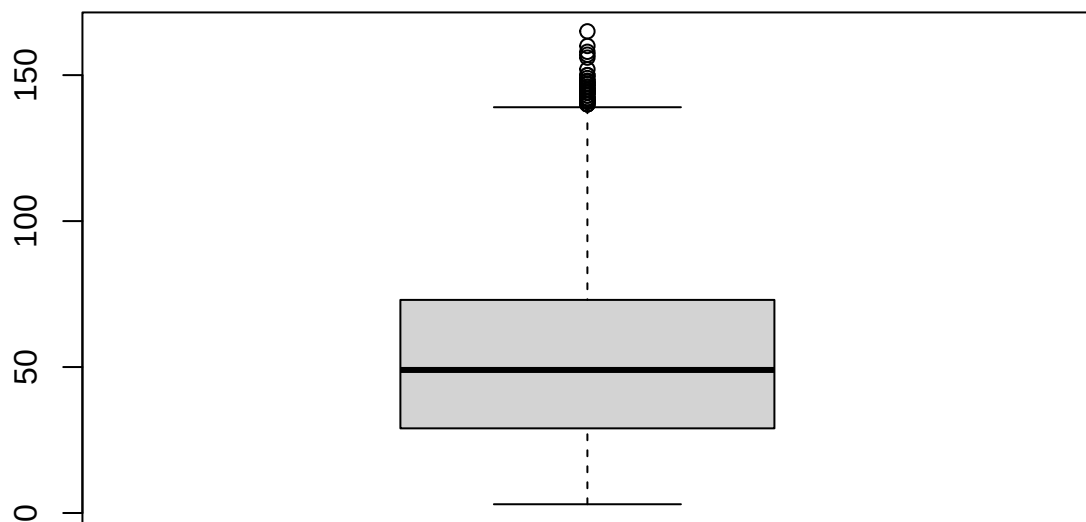


Boxplots

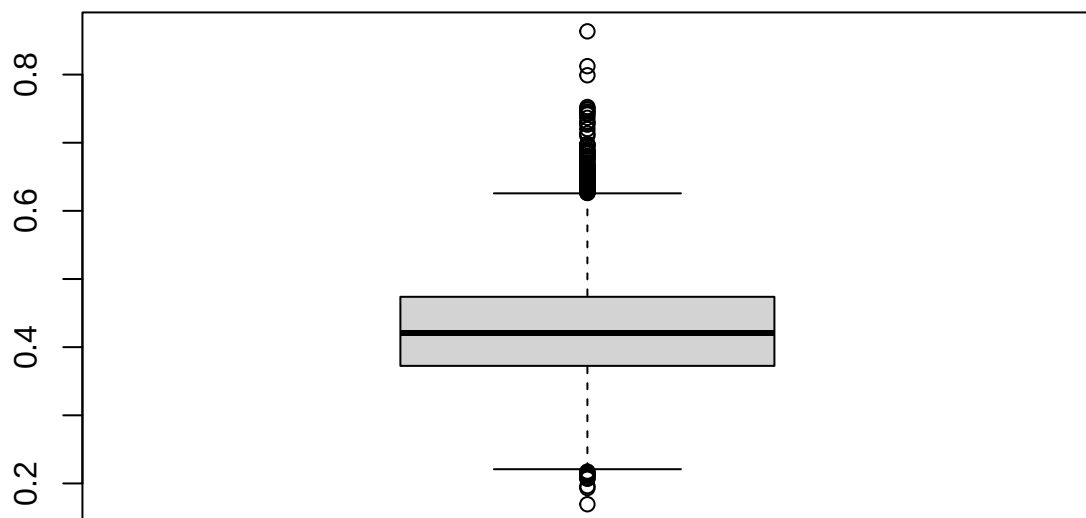
```
boxplot(BattingDataSlim$HR)
```

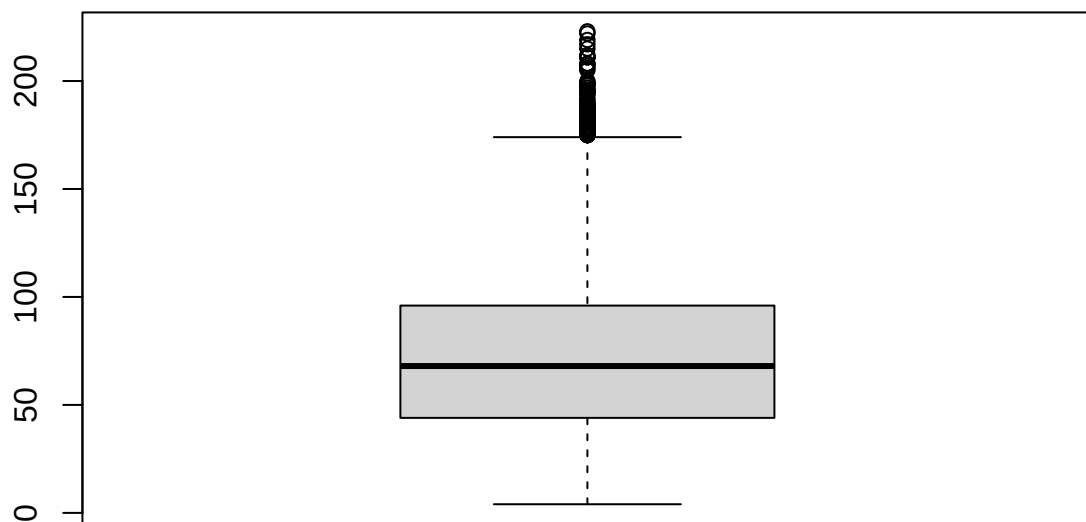
```
boxplot(BattingDataSlim$RBI)
```



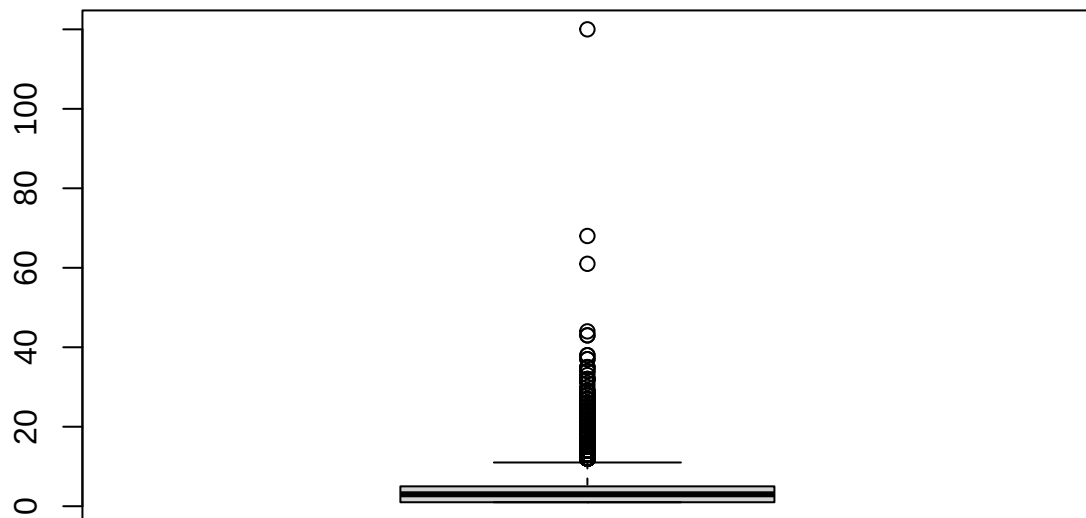
```
boxplot(BattingDataSlim$SLG)
```



```
boxplot(BattingDataSlim$S0)
```

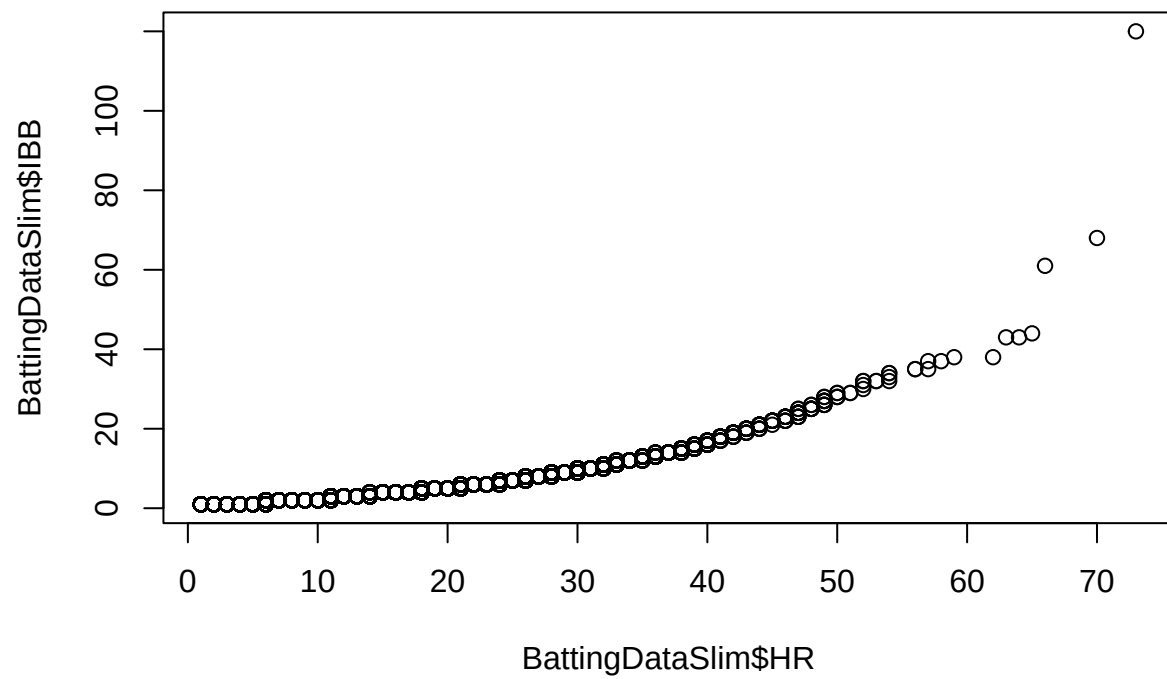


```
boxplot(BattingDataSlim$IBB)
```

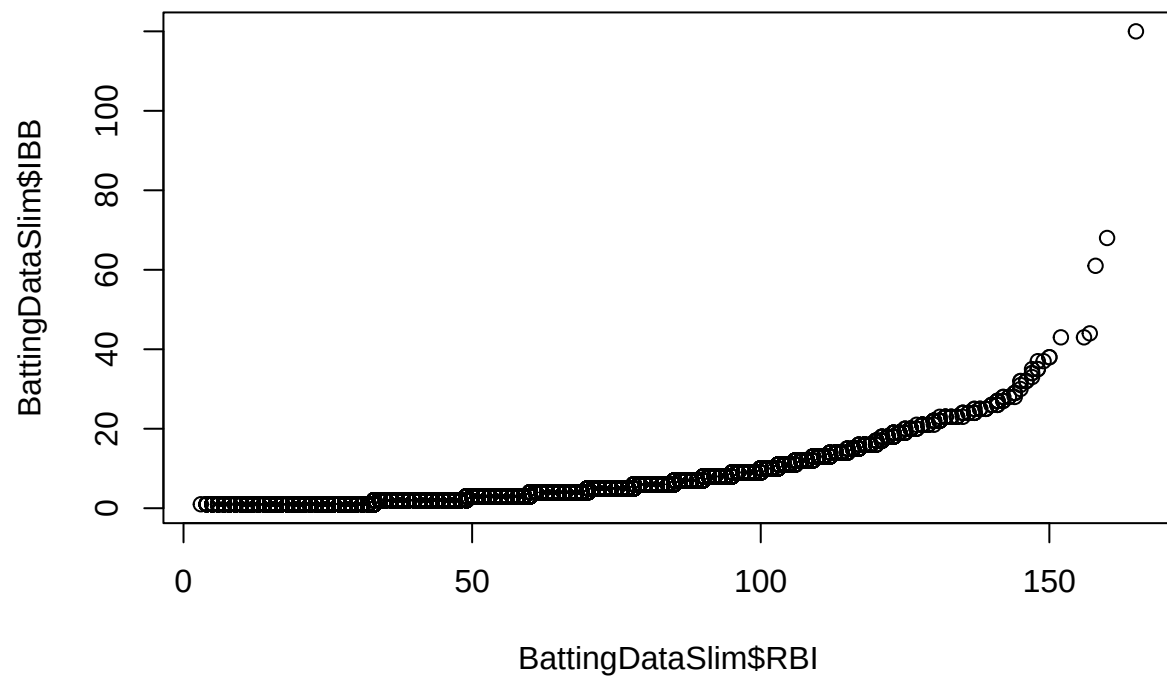


QQ Plots

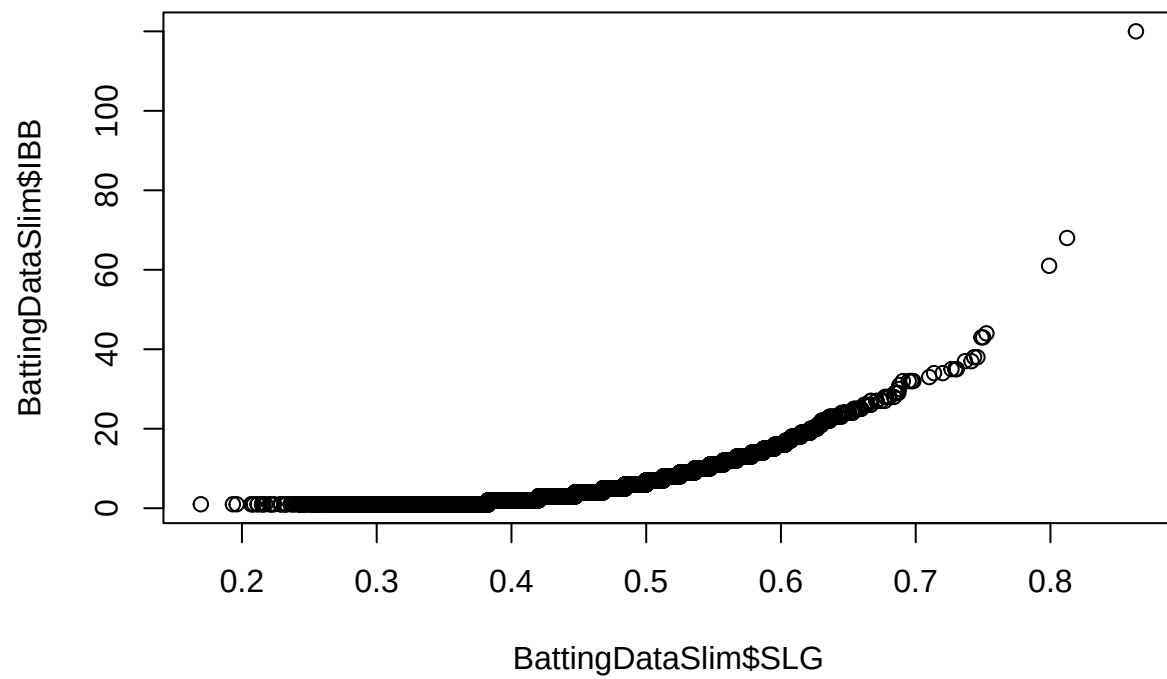
```
qqplot(BattingDataSlim$HR, BattingDataSlim$IBB)
```



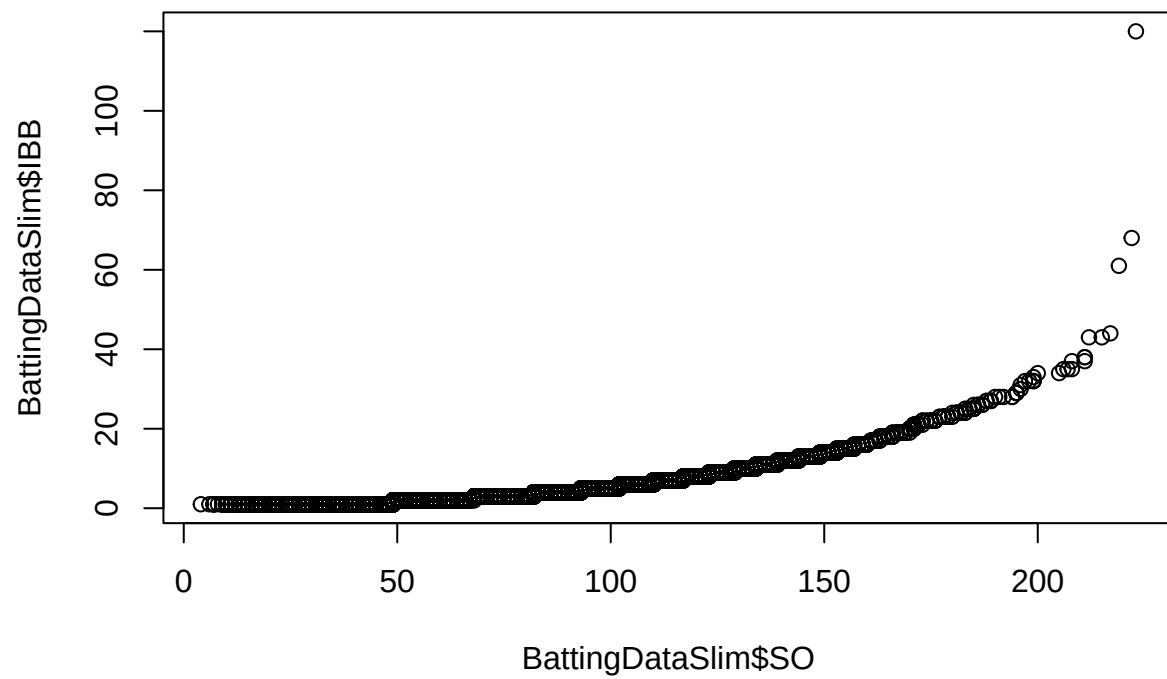
```
qqplot(BattingDataSlim$RBI, BattingDataSlim$IBB)
```



```
qqplot(BattingDataSlim$SLG, BattingDataSlim$IBB)
```

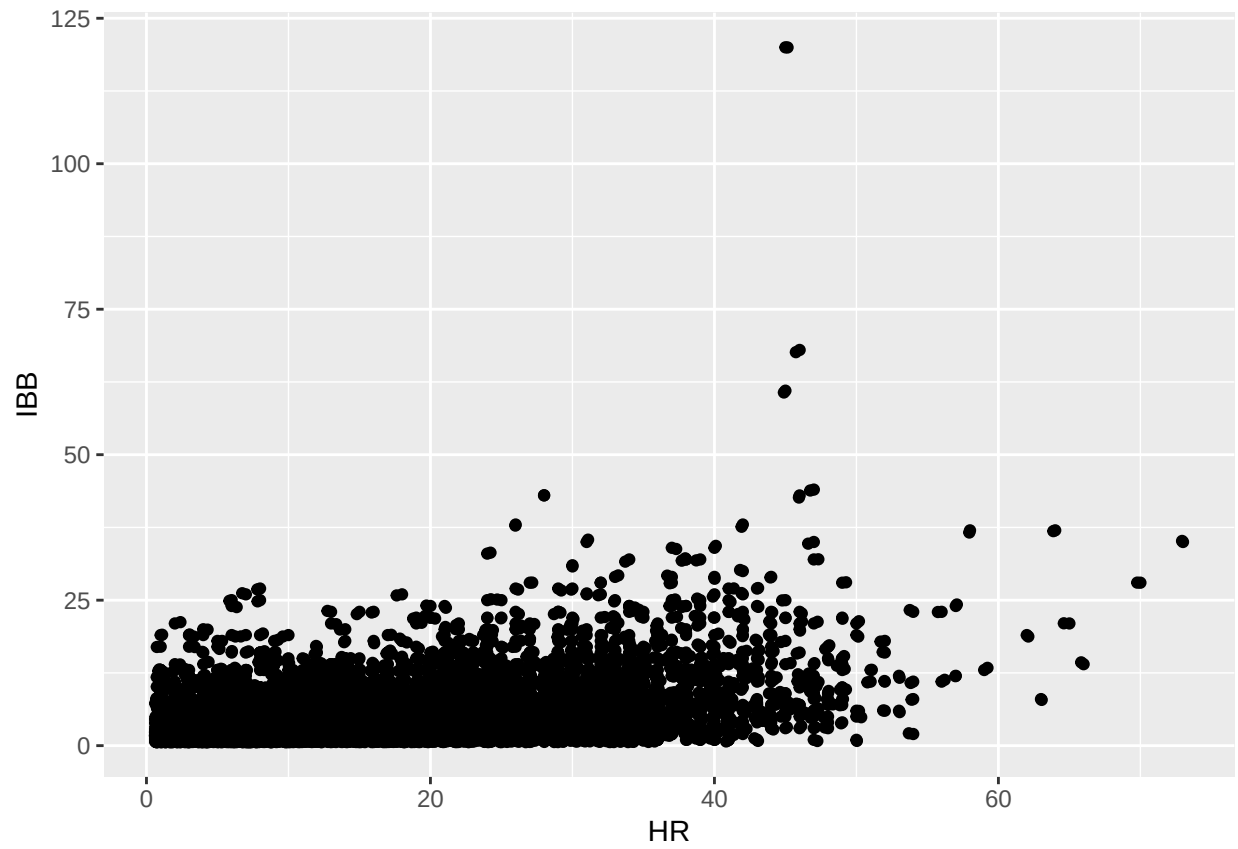


```
qqplot(BattingDataSlim$S0, BattingDataSlim$IBB)
```

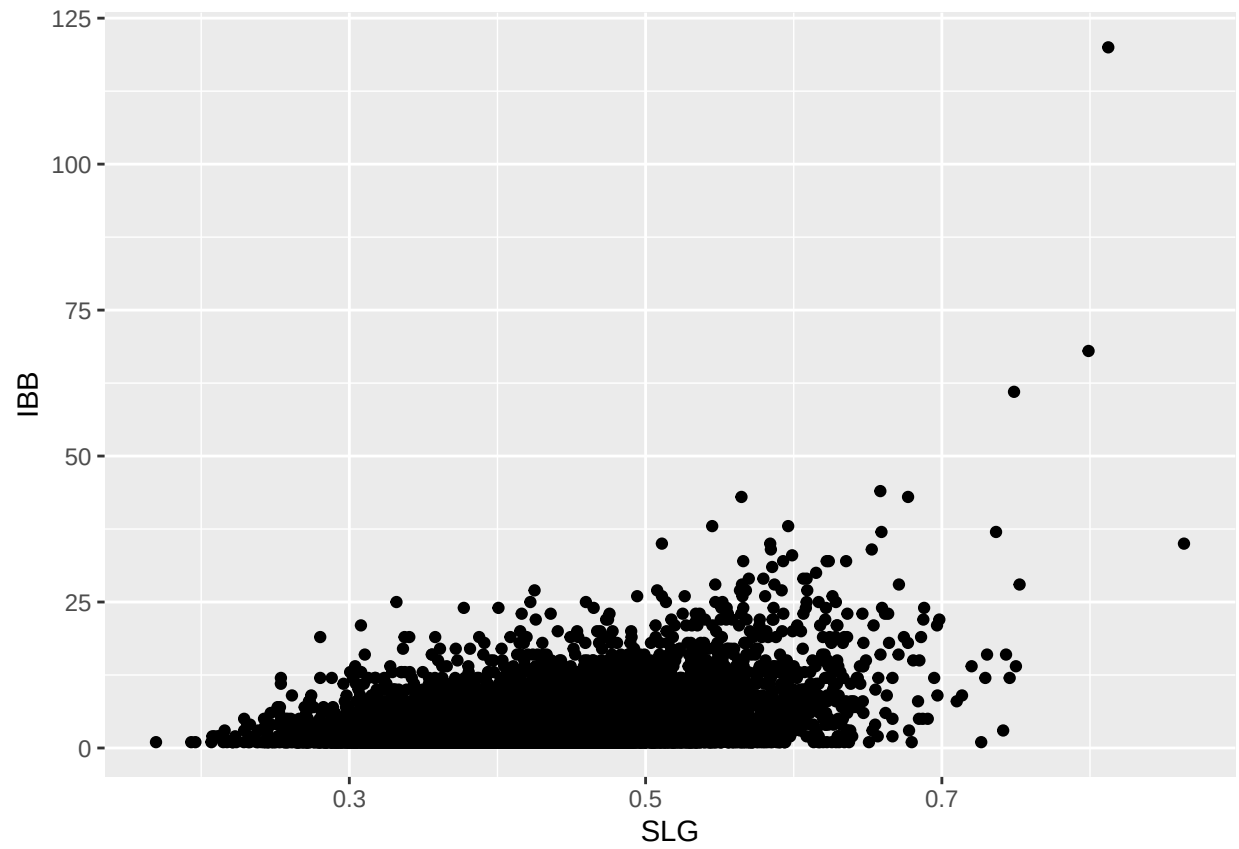



Scatter Plots with Regression Lines (Question 3)

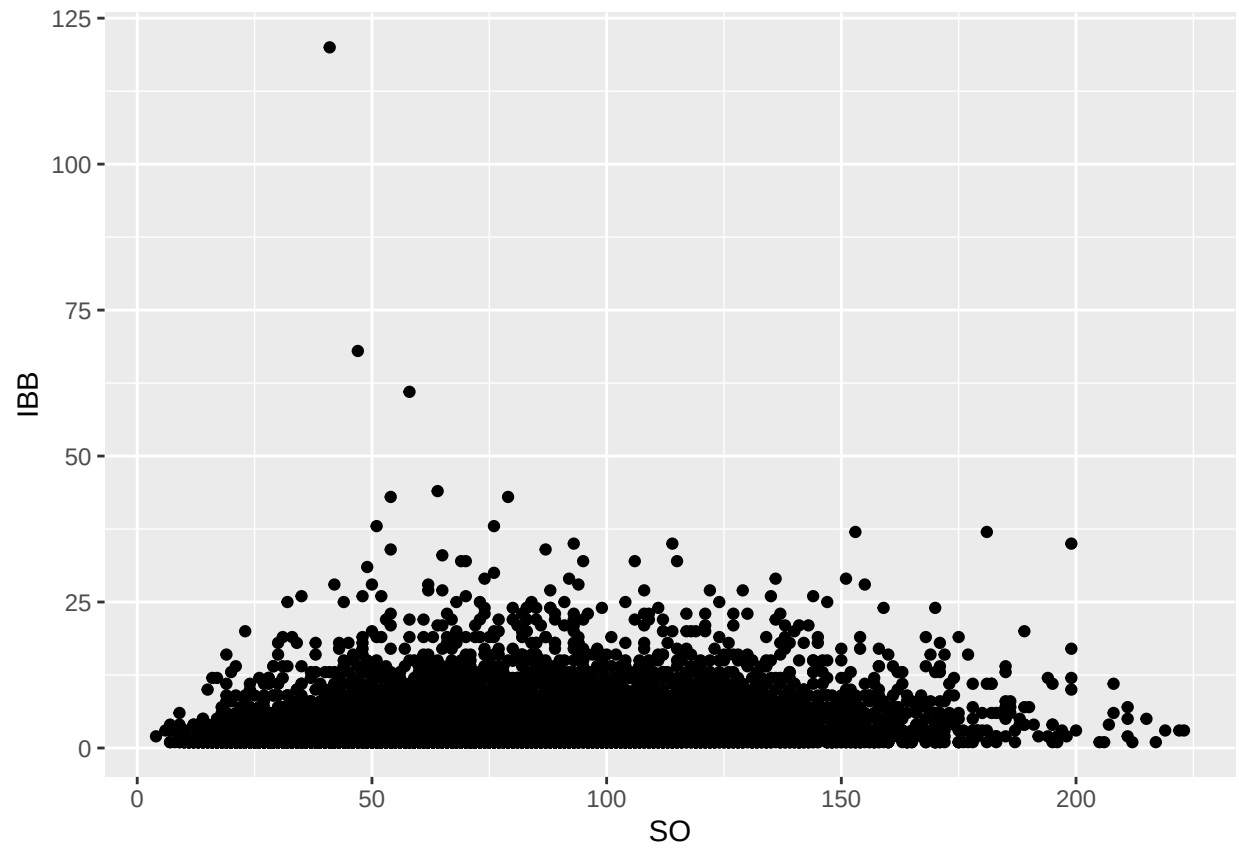
```
ggplot(BattingDataSlim, aes(x = HR, y = IBB)) +  
  geom_point() +  
  geom_jitter()
```



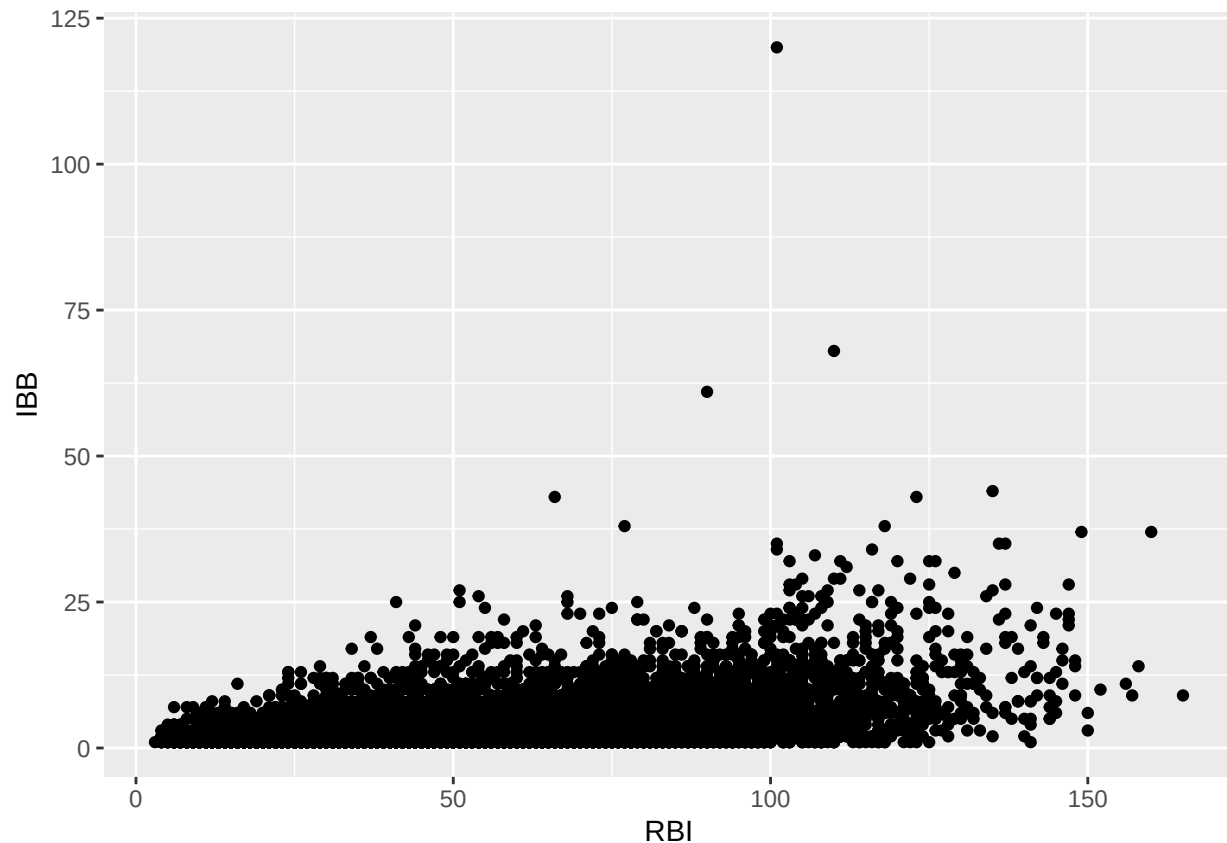
```
ggplot(BattingDataSlim, aes(x = SLG, y = IBB)) +  
  geom_point()
```



```
ggplot(BattingDataSlim, aes(x = S0, y = IBB)) +  
  geom_point()
```



```
ggplot(BattingDataSlim, aes(x = RBI, y = IBB)) +  
  geom_point()
```



Checking for Transformations (Question 4)

```
summary(lm(BattingDataSlim$IBB ~ (BattingDataSlim$HR))) #R^2 0.2063
```

```
##
## Call:
## lm(formula = BattingDataSlim$IBB ~ (BattingDataSlim$HR))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.838  -1.916  -0.784   0.992  110.105
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.406768   0.060087   23.41  <2e-16 ***
## BattingDataSlim$HR 0.188619   0.003573   52.78  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.81 on 10715 degrees of freedom
## Multiple R-squared:  0.2064, Adjusted R-squared:  0.2063
## F-statistic: 2786 on 1 and 10715 DF, p-value: < 2.2e-16
```

```
summary(lm(BattingDataSlim$IBB ~ exp(BattingDataSlim$HR))) #R^2 0.005221
```

```
##
## Call:
## lm(formula = BattingDataSlim$IBB ~ exp(BattingDataSlim$HR))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.911  -2.911  -0.911   1.089  116.089
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3.911e+00  4.120e-02  94.922 < 2e-16 ***
## exp(BattingDataSlim$HR) 6.379e-31  8.431e-32   7.566 4.18e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.265 on 10715 degrees of freedom
## Multiple R-squared:  0.005313, Adjusted R-squared:  0.005221
## F-statistic: 57.24 on 1 and 10715 DF, p-value: 4.18e-14
```

```
summary(lm(BattingDataSlim$IBB ~ sqrt(BattingDataSlim$HR))) #R^2 0.1718
```

```
##
## Call:
## lm(formula = BattingDataSlim$IBB ~ sqrt(BattingDataSlim$HR))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.599  -2.170  -0.756   1.020  111.860
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -0.34078    0.09773  -3.487 0.000491 ***
## sqrt(BattingDataSlim$HR) 1.26425    0.02681  47.162 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.891 on 10715 degrees of freedom
## Multiple R-squared:  0.1719, Adjusted R-squared:  0.1718
## F-statistic: 2224 on 1 and 10715 DF, p-value: < 2.2e-16
```

```
summary(lm(BattingDataSlim$IBB ~ log(BattingDataSlim$HR))) #R^2 0.1238
```

```
##
## Call:
## lm(formula = BattingDataSlim$IBB ~ log(BattingDataSlim$HR))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.620  -2.326  -0.919   0.967  113.549
```

```
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      0.33151    0.09982   3.321   9e-04 ***
## log(BattingDataSlim$HR) 1.60755    0.04130  38.927  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.003 on 10715 degrees of freedom
## Multiple R-squared:  0.1239, Adjusted R-squared:  0.1238
## F-statistic: 1515 on 1 and 10715 DF, p-value: < 2.2e-16
```

```
summary(lm(BattingDataSlim$IBB ~ (BattingDataSlim$RBI))) #R^2 0.2305
```

```
##
## Call:
## lm(formula = BattingDataSlim$IBB ~ (BattingDataSlim$RBI))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.075  -1.964  -0.559   0.999  112.735
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      0.171088    0.075332   2.271   0.0232 *
## BattingDataSlim$RBI 0.070239    0.001239  56.670  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.751 on 10715 degrees of freedom
## Multiple R-squared:  0.2306, Adjusted R-squared:  0.2305
## F-statistic: 3212 on 1 and 10715 DF, p-value: < 2.2e-16
```

```
summary(lm(BattingDataSlim$IBB ~ exp(BattingDataSlim$RBI))) #R^2 5.118e-05
```

```
##
## Call:
## lm(formula = BattingDataSlim$IBB ~ exp(BattingDataSlim$RBI))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.913  -2.913  -0.913   1.087  116.087
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      3.913e+00  4.131e-02  94.742  <2e-16 ***
## exp(BattingDataSlim$RBI) 1.168e-71  9.386e-72   1.244   0.213
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.276 on 10715 degrees of freedom
## Multiple R-squared:  0.0001445, Adjusted R-squared:  5.118e-05
## F-statistic: 1.548 on 1 and 10715 DF, p-value: 0.2134
```

```
summary(lm(BattingDataSlim$IBB ~ sqrt(BattingDataSlim$RBI))) #R^2 0.2057
```

```
##
## Call:
## lm(formula = BattingDataSlim$IBB ~ sqrt(BattingDataSlim$RBI))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.529  -2.114  -0.615   1.058  113.201
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -2.7306     0.1314  -20.79  <2e-16 ***
## sqrt(BattingDataSlim$RBI)  0.9482     0.0180   52.69  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.811 on 10715 degrees of freedom
## Multiple R-squared:  0.2058, Adjusted R-squared:  0.2057
## F-statistic: 2776 on 1 and 10715 DF, p-value: < 2.2e-16
```

```
summary(lm(BattingDataSlim$IBB ~ log(BattingDataSlim$RBI))) #R^2 0.1707
```

```
##
## Call:
## lm(formula = BattingDataSlim$IBB ~ log(BattingDataSlim$RBI))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.111  -2.234  -0.734   1.107  113.817
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -6.65820     0.22817  -29.18  <2e-16 ***
## log(BattingDataSlim$RBI)  2.78243     0.05923   46.98  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.894 on 10715 degrees of freedom
## Multiple R-squared:  0.1708, Adjusted R-squared:  0.1707
## F-statistic: 2207 on 1 and 10715 DF, p-value: < 2.2e-16
```

```
summary(lm(BattingDataSlim$IBB ~ (BattingDataSlim$SLG))) #R^2 0.1487
```

```
##
## Call:
## lm(formula = BattingDataSlim$IBB ~ (BattingDataSlim$SLG))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.362  -2.248  -0.824   1.123  107.798
```



```
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      -5.2222     0.2146  -24.34  <2e-16 ***
## BattingDataSlim$SLG  21.4491     0.4957   43.27  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.945 on 10715 degrees of freedom
## Multiple R-squared:  0.1488, Adjusted R-squared:  0.1487
## F-statistic: 1872 on 1 and 10715 DF, p-value: < 2.2e-16

summary(lm(BattingDataSlim$IBB ~ exp(BattingDataSlim$SLG))) #R^2 0.1601
```

```
##
## Call:
## lm(formula = BattingDataSlim$IBB ~ exp(BattingDataSlim$SLG))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.494  -2.209  -0.806   1.123  105.869
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      -17.9528     0.4851  -37.01  <2e-16 ***
## exp(BattingDataSlim$SLG)  14.2397     0.3150   45.21  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.919 on 10715 degrees of freedom
## Multiple R-squared:  0.1602, Adjusted R-squared:  0.1601
## F-statistic: 2044 on 1 and 10715 DF, p-value: < 2.2e-16
```

```
summary(lm(BattingDataSlim$IBB ~ sqrt(BattingDataSlim$SLG))) #R^2 0.1365
```

```
##
## Call:
## lm(formula = BattingDataSlim$IBB ~ sqrt(BattingDataSlim$SLG))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
##  -8.362  -2.291  -0.861   1.132  109.321
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      -13.5839     0.4267  -31.84  <2e-16 ***
## sqrt(BattingDataSlim$SLG)  26.9197     0.6537   41.18  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.973 on 10715 degrees of freedom
## Multiple R-squared:  0.1366, Adjusted R-squared:  0.1365
## F-statistic: 1696 on 1 and 10715 DF, p-value: < 2.2e-16
```

```
summary(lm(BattingDataSlim$IBB ~ log(BattingDataSlim$SLG))) #R^2 0.1244
```

```
##
## Call:
## lm(formula = BattingDataSlim$IBB ~ log(BattingDataSlim$SLG))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.484  -2.331  -0.910   1.126  110.590
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      11.1359     0.1890   58.92  <2e-16 ***
## log(BattingDataSlim$SLG)  8.3032     0.2127   39.03  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.001 on 10715 degrees of freedom
## Multiple R-squared:  0.1245, Adjusted R-squared:  0.1244
## F-statistic: 1524 on 1 and 10715 DF, p-value: < 2.2e-16
```

```
summary(lm(BattingDataSlim$IBB ~ (BattingDataSlim$S0))) #R^2 0.04826
```

```
##
## Call:
## lm(formula = BattingDataSlim$IBB ~ (BattingDataSlim$S0))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.629  -2.242  -1.164   0.862  116.913
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      2.028706     0.090289   22.47  <2e-16 ***
## BattingDataSlim$S0 0.025807     0.001106   23.33  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.172 on 10715 degrees of freedom
## Multiple R-squared:  0.04835, Adjusted R-squared:  0.04826
## F-statistic: 544.4 on 1 and 10715 DF, p-value: < 2.2e-16
```

```
summary(lm(BattingDataSlim$IBB ~ exp(BattingDataSlim$S0))) #R^2 -8.603e-05
```

```
##
## Call:
## lm(formula = BattingDataSlim$IBB ~ exp(BattingDataSlim$S0))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.914  -2.914  -0.914   1.086  116.086
```

```
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      3.914e+00  4.131e-02   94.75  <2e-16 ***
## exp(BattingDataSlim$S0) -1.593e-97  5.699e-97   -0.28    0.78
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.276 on 10715 degrees of freedom
## Multiple R-squared:  7.293e-06, Adjusted R-squared:  -8.603e-05
## F-statistic: 0.07815 on 1 and 10715 DF, p-value: 0.7798
```

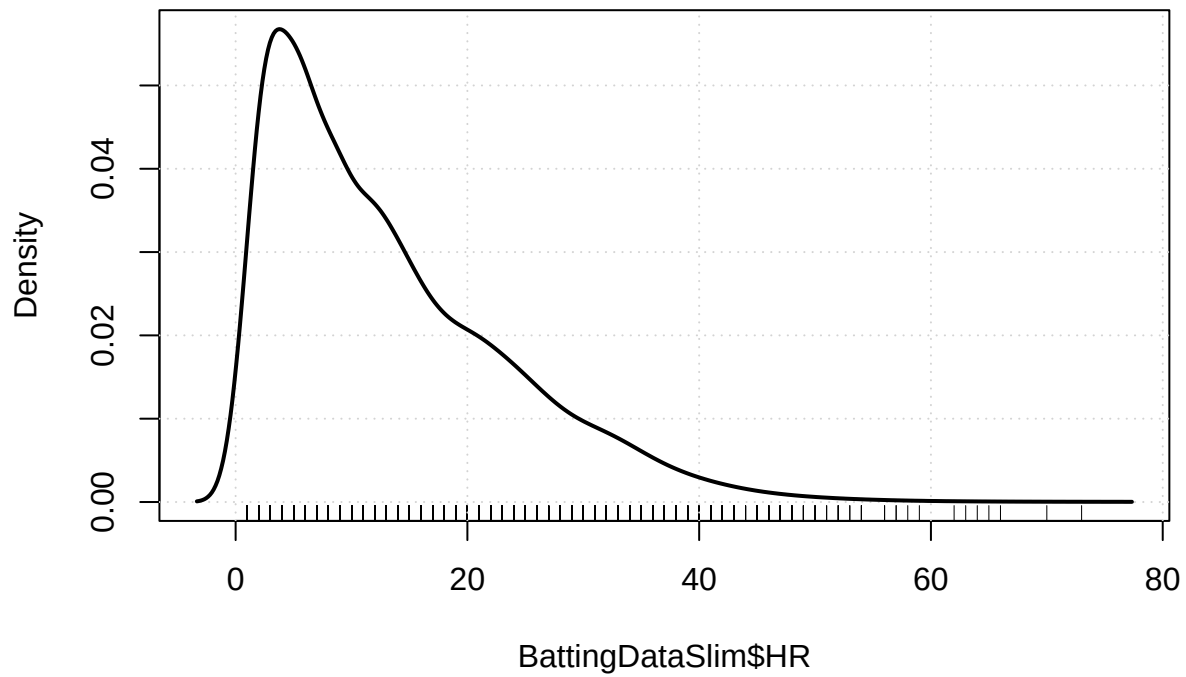
```
summary(lm(BattingDataSlim$IBB ~ sqrt(BattingDataSlim$S0))) #R^2 0.05132
```

```
##
## Call:
## lm(formula = BattingDataSlim$IBB ~ sqrt(BattingDataSlim$S0))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.830  -2.307  -1.118   0.861  116.931
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      0.17760    0.16019   1.109   0.268
## sqrt(BattingDataSlim$S0) 0.45163    0.01874  24.098  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.165 on 10715 degrees of freedom
## Multiple R-squared:  0.05141, Adjusted R-squared:  0.05132
## F-statistic: 580.7 on 1 and 10715 DF, p-value: < 2.2e-16
```

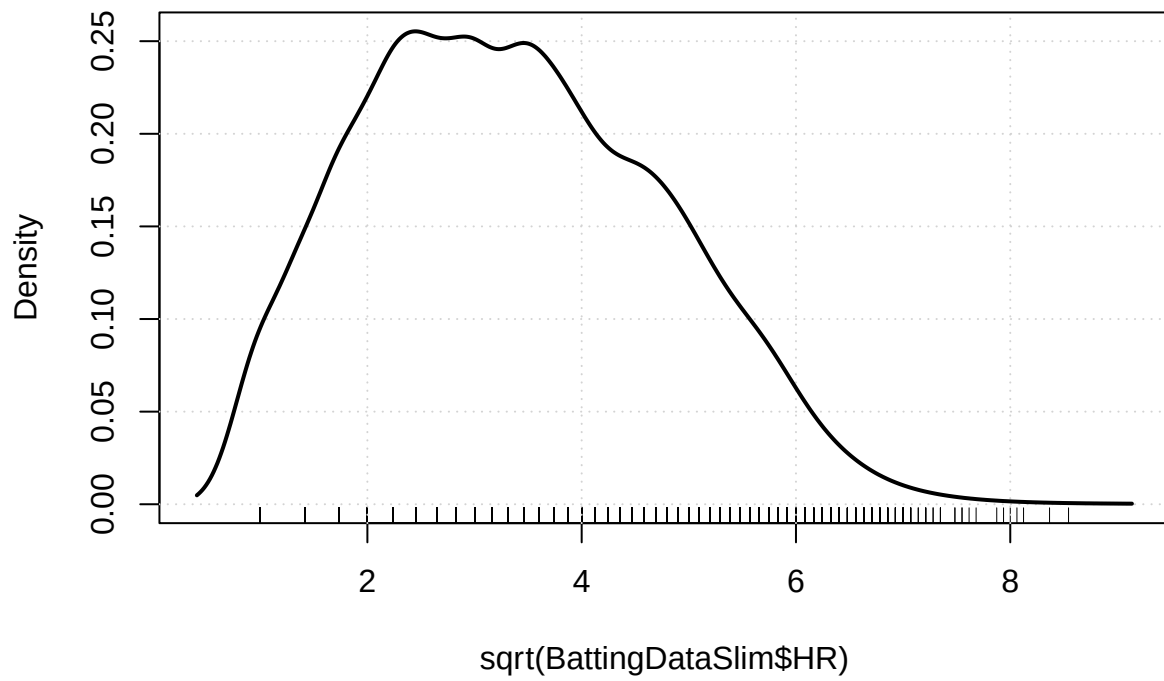
```
summary(lm(BattingDataSlim$IBB ~ log(BattingDataSlim$S0))) #R^2 0.05137
```

```
##
## Call:
## lm(formula = BattingDataSlim$IBB ~ log(BattingDataSlim$S0))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.070  -2.342  -1.097   0.900  116.860
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      -3.39005    0.30559  -11.09  <2e-16 ***
## log(BattingDataSlim$S0) 1.75845    0.07293   24.11  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.165 on 10715 degrees of freedom
## Multiple R-squared:  0.05146, Adjusted R-squared:  0.05137
## F-statistic: 581.3 on 1 and 10715 DF, p-value: < 2.2e-16
```

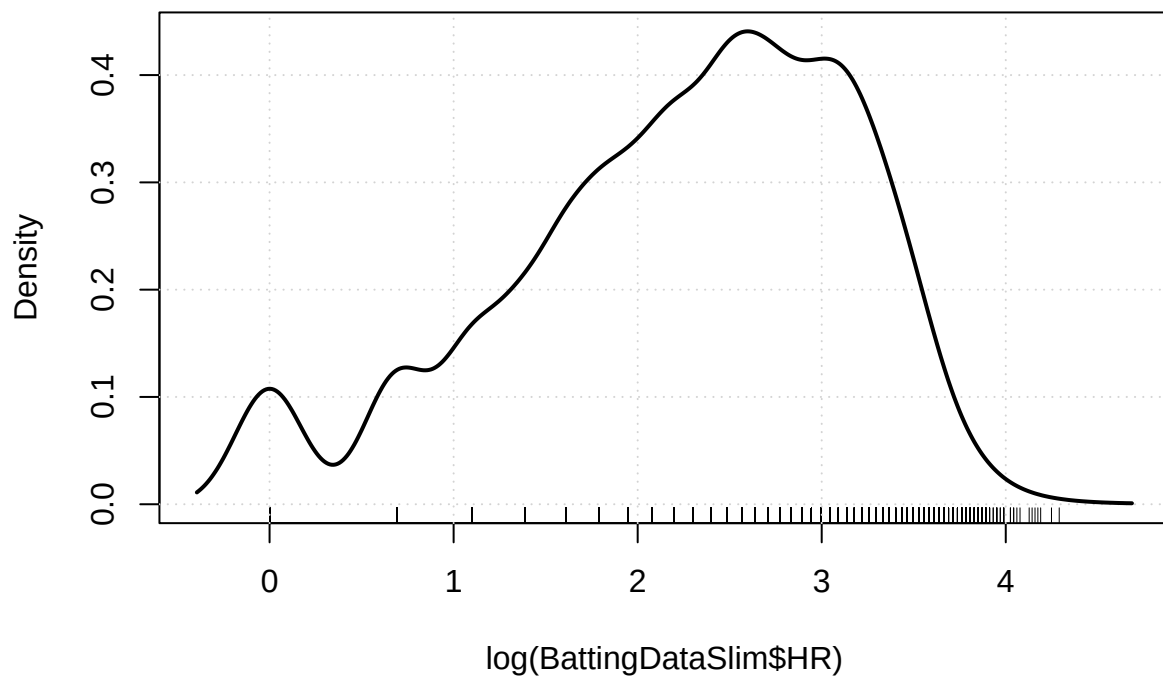
```
densityPlot(BattingDataSlim$HR) #best R^2, will use this for the Transformation
```



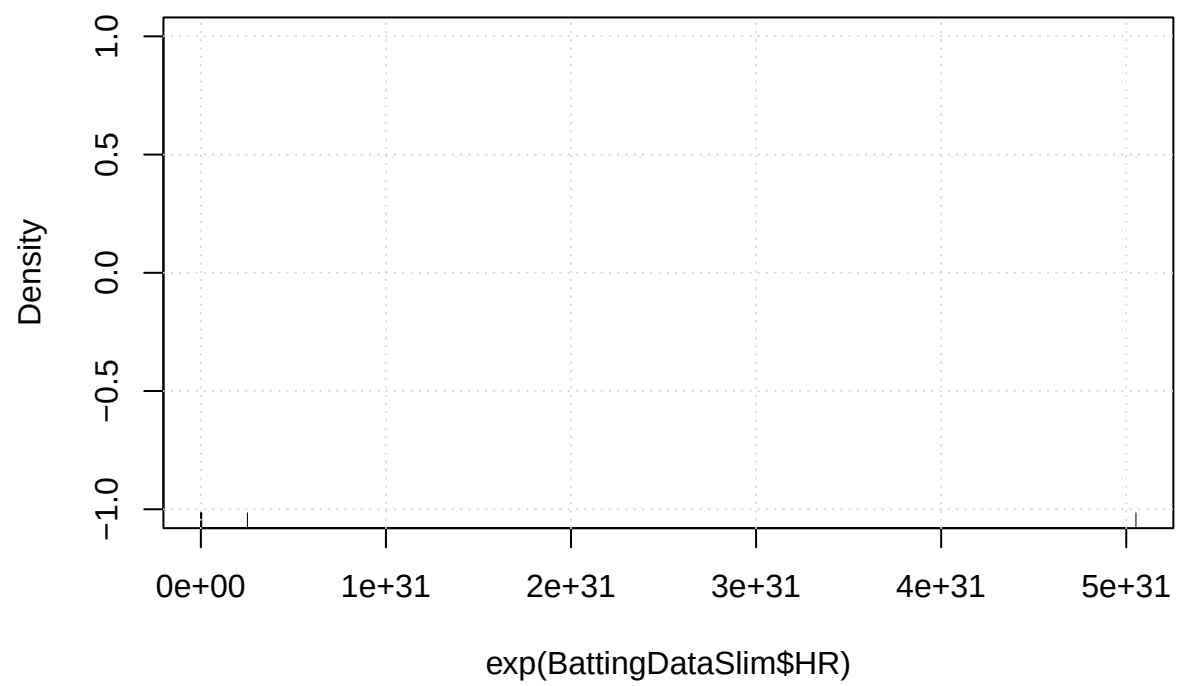
```
densityPlot(sqrt(BattingDataSlim$HR)) #best visually
```



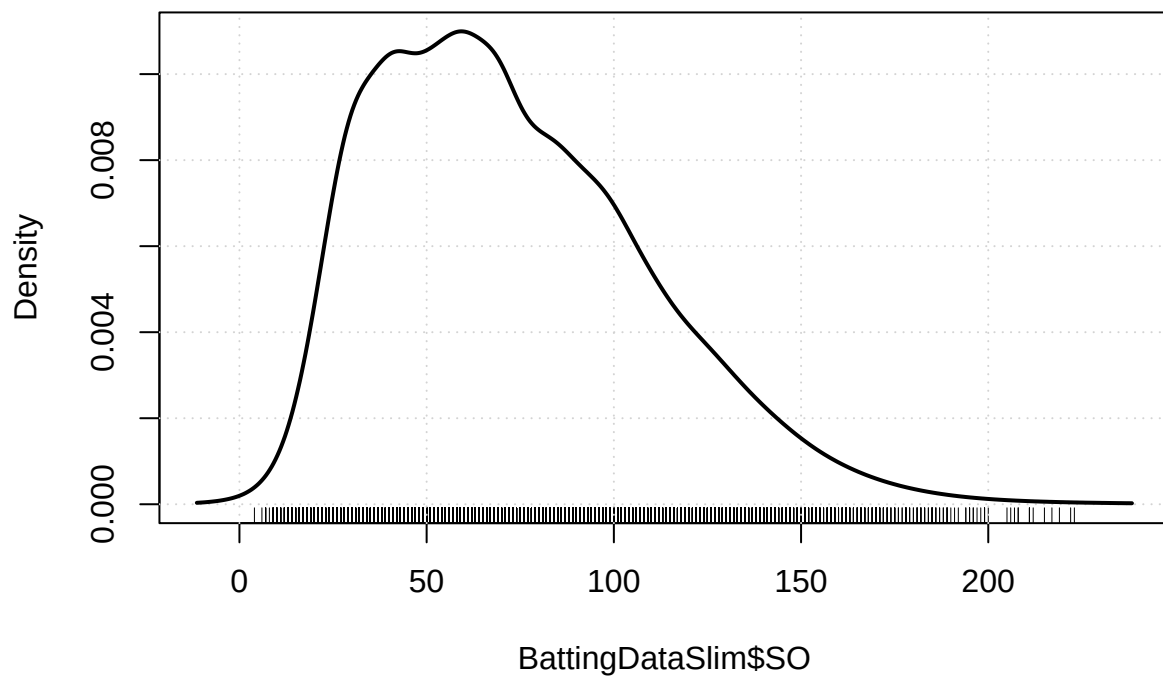
```
densityPlot(log(BattingDataSlim$HR))
```



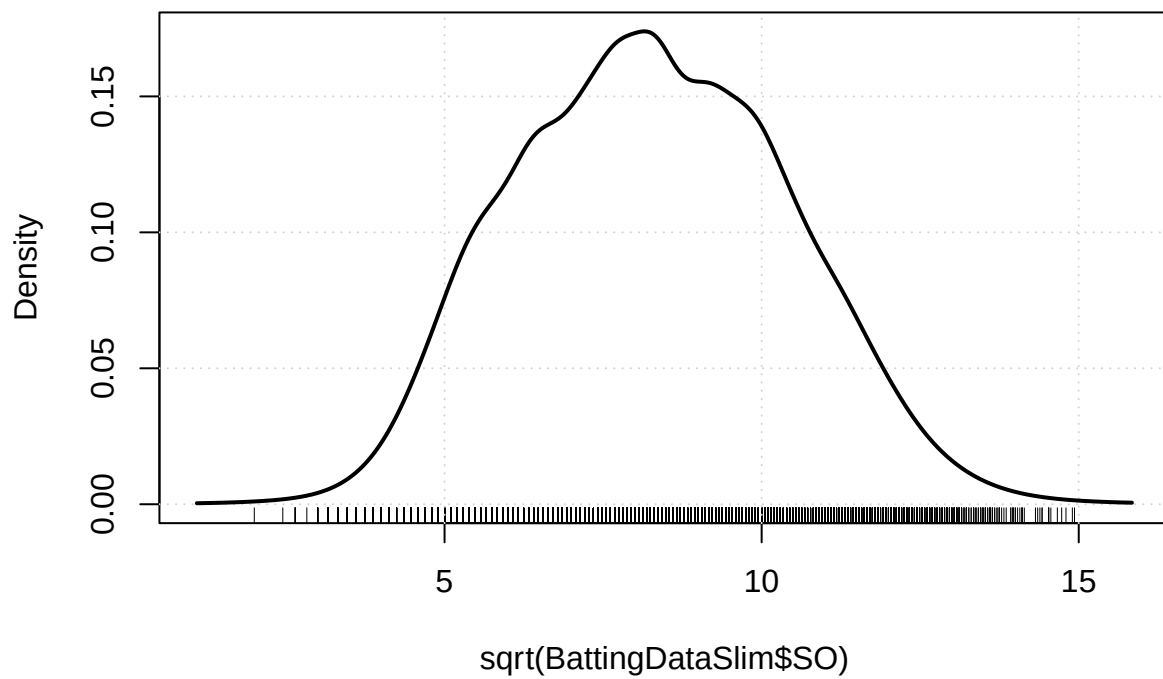
```
densityPlot(exp(BattingDataSlim$HR))
```



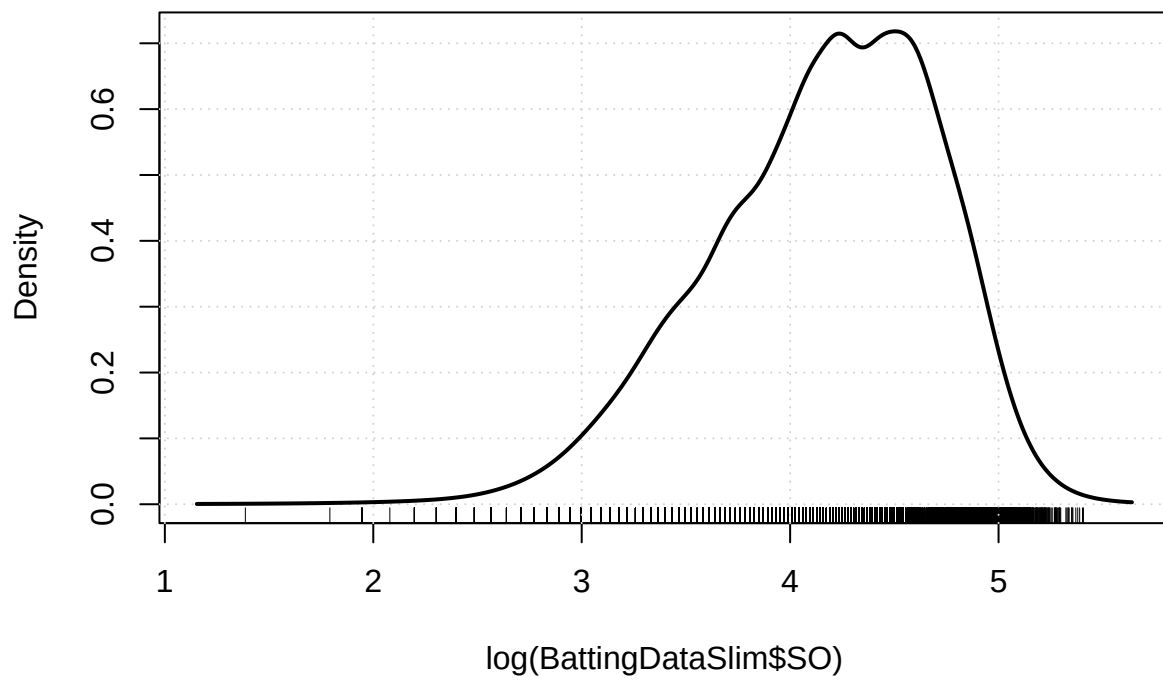
```
densityPlot(BattingDataSlim$S0) #best visually
```



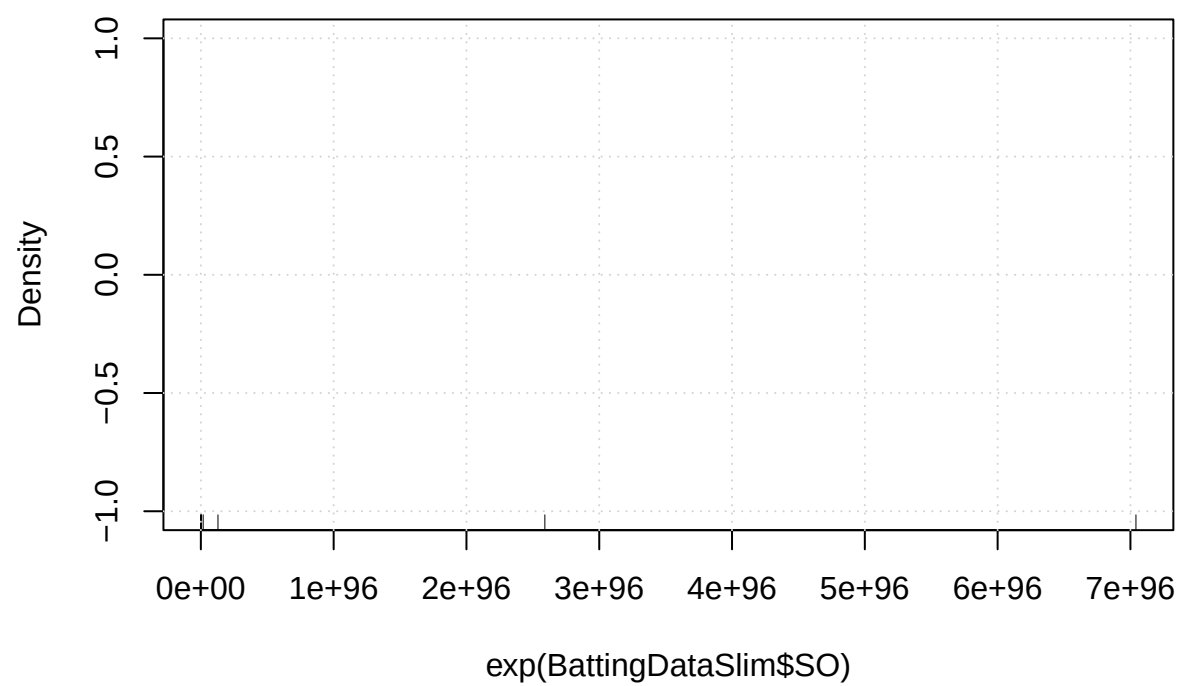
```
densityPlot(sqrt(BattingDataSlim$SO)) #best R2, will use this for the Transformation
```

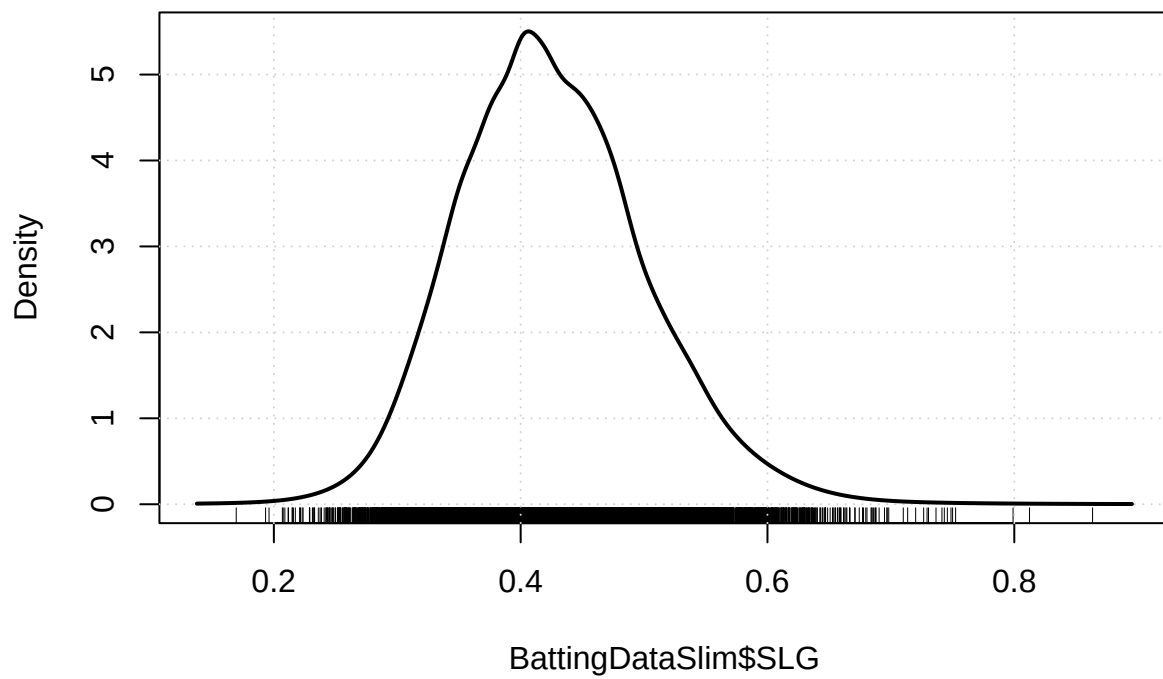
```
densityPlot(log(BattingDataSlim$SO))
```



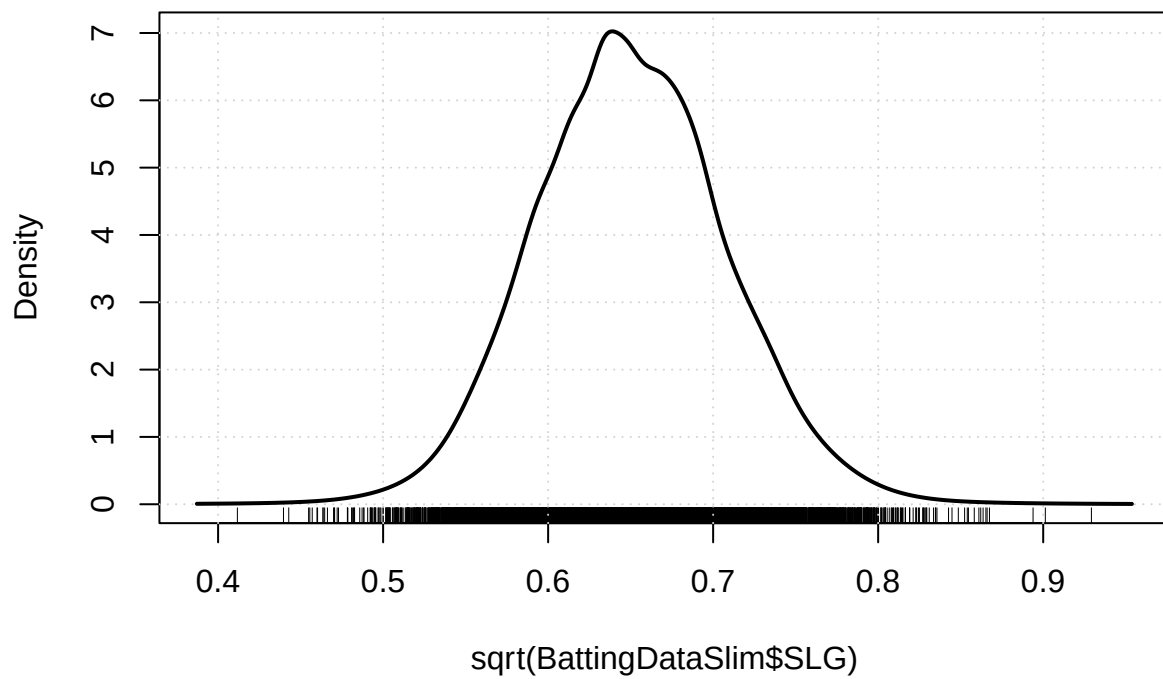
```
densityPlot(exp(BattingDataSlim$SO))
```



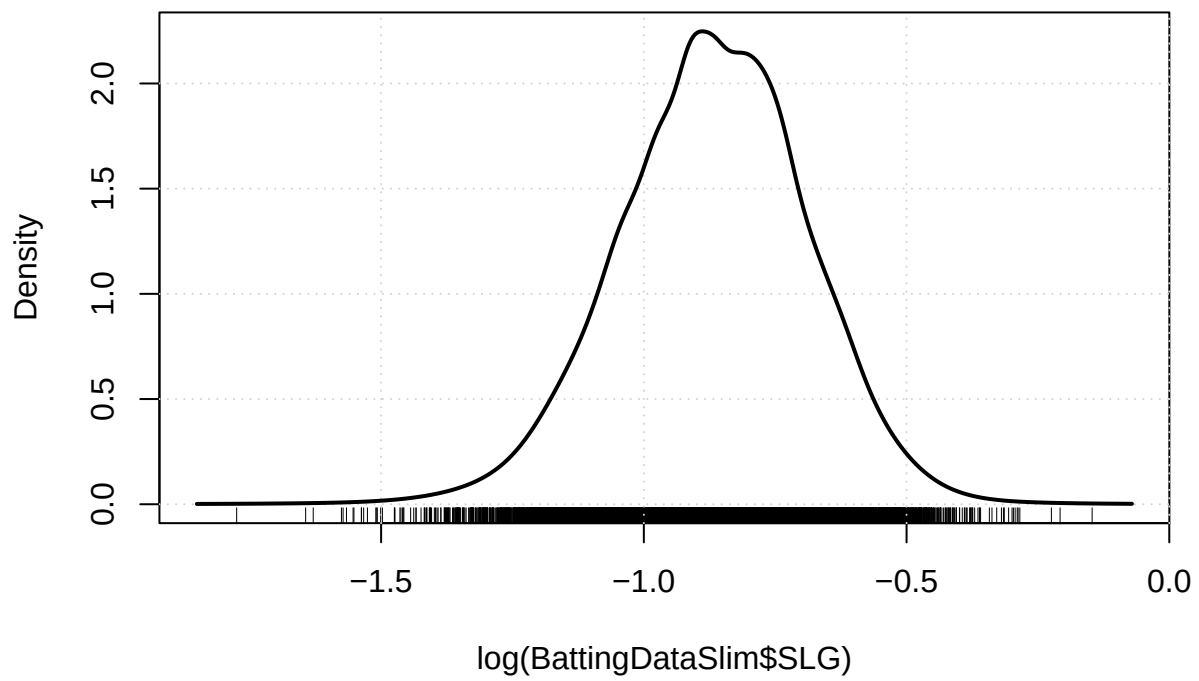
```
densityPlot(BattingDataSlim$SLG) #best visually
```



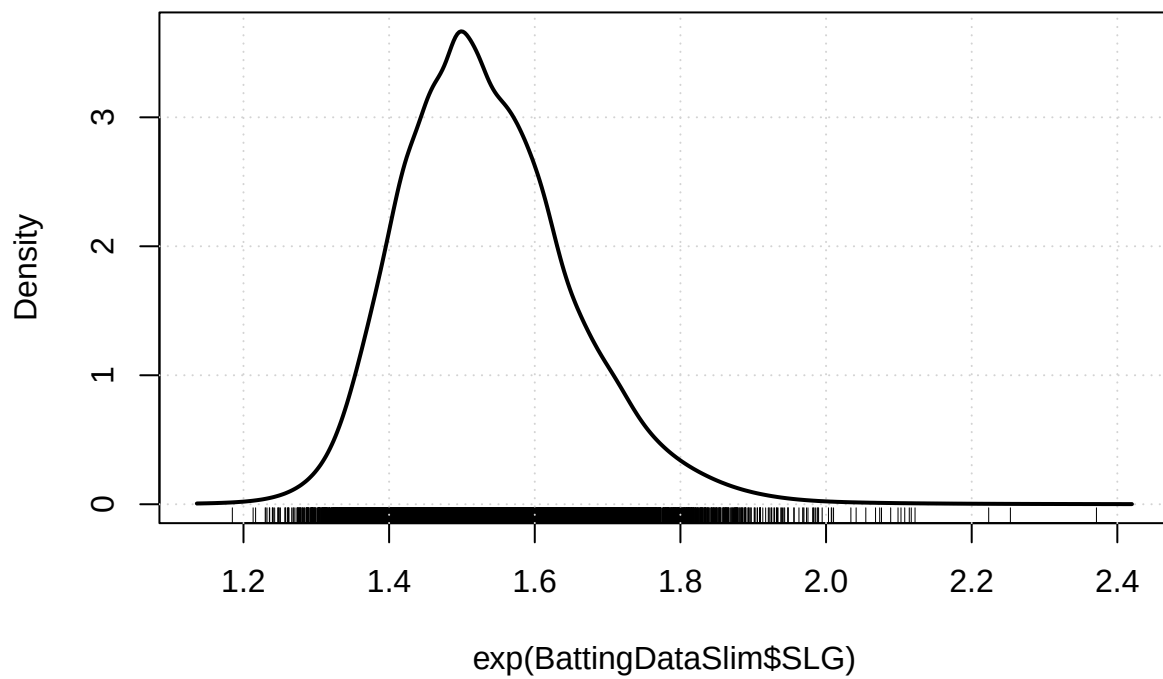
```
densityPlot(sqrt(BattingDataSlim$SLG))
```



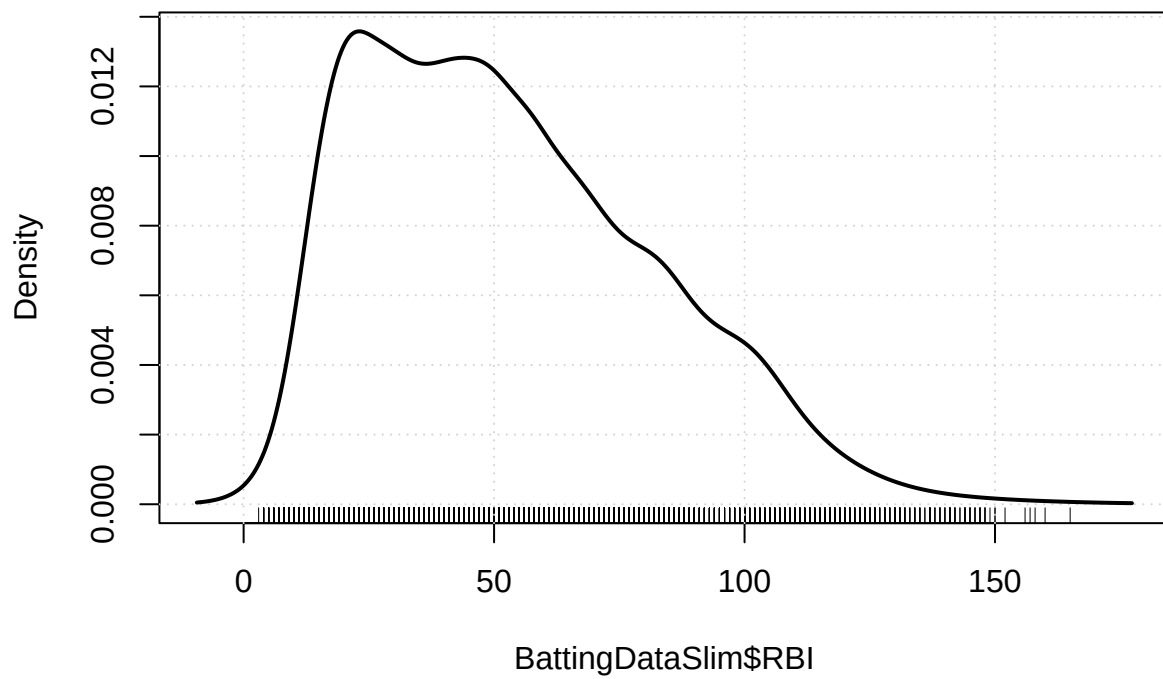
```
densityPlot(log(BattingDataSlim$SLG))
```



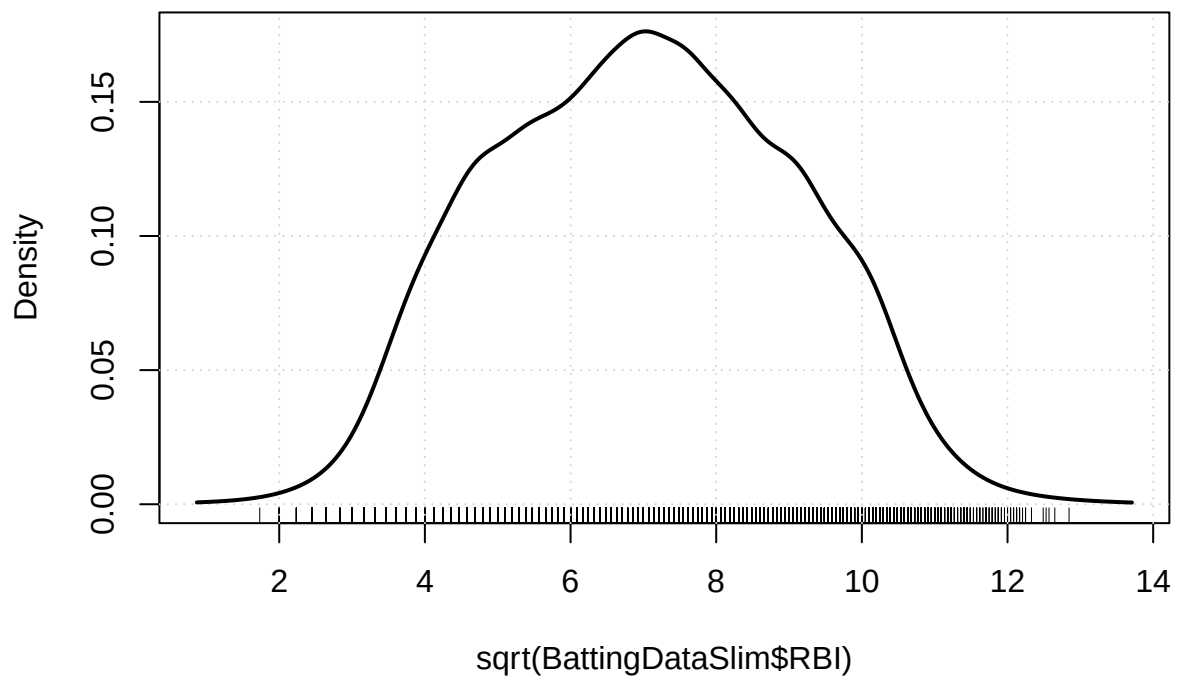
```
densityPlot(exp(BattingDataSlim$SLG)) #best  $R^2$ , will use this for the Transformation
```



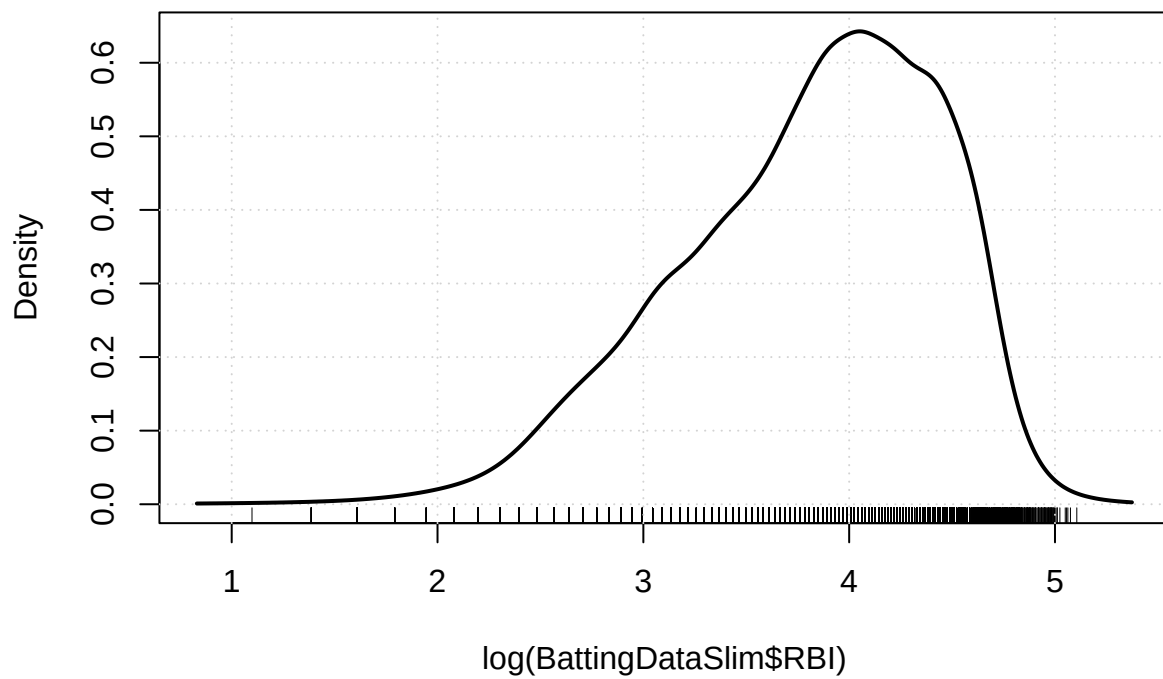
```
densityPlot(BattingDataSlim$RBI) #best visually, best R2, will use this for the Transformation
```



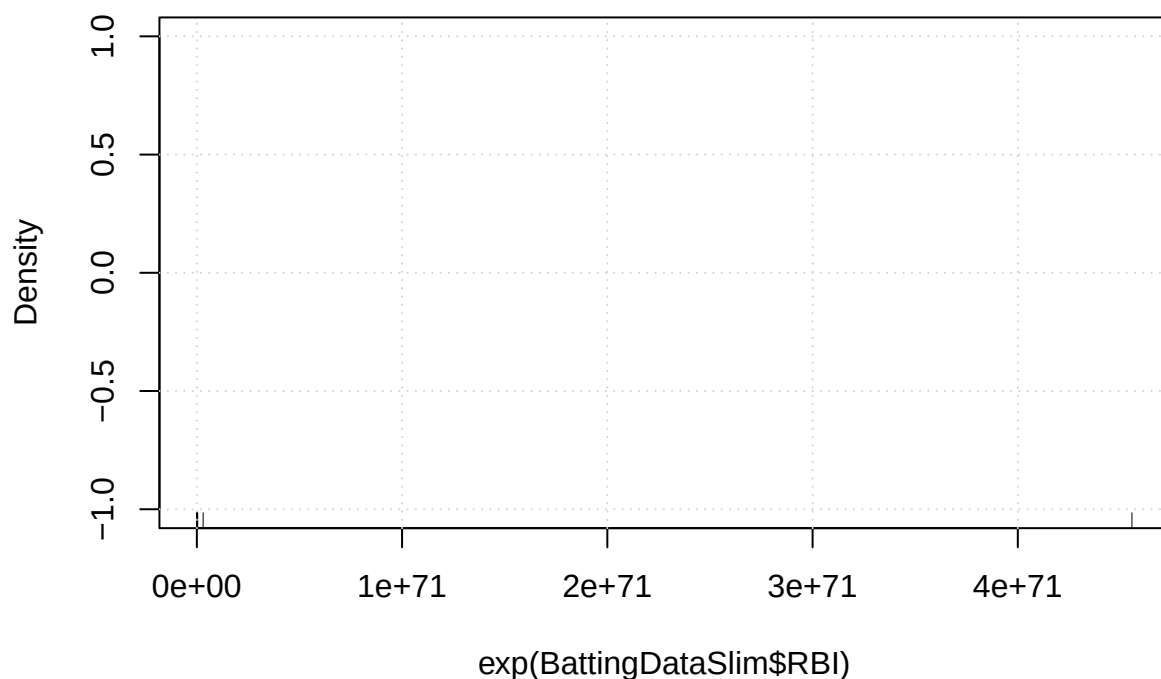
```
densityPlot(sqrt(BattingDataSlim$RBI))
```

```
densityPlot(log(BattingDataSlim$RBI))
```



```
densityPlot(exp(BattingDataSlim$RBI))
```



Post-Transformation Linear Models (Question 6)

```
model1 <- lm(BattingDataSlim$IBB ~ (BattingDataSlim$HR))
model2 <- lm(BattingDataSlim$IBB ~ sqrt(BattingDataSlim$SO))
model3 <- lm(BattingDataSlim$IBB ~ exp(BattingDataSlim$SLG))
model4 <- lm(BattingDataSlim$IBB ~ (BattingDataSlim$RBI))
```

```
summary(model1)
```

```
##
## Call:
## lm(formula = BattingDataSlim$IBB ~ (BattingDataSlim$HR))
##
## Residuals:
```

##	Min	1Q	Median	3Q	Max
##	-9.838	-1.916	-0.784	0.992	110.105

```
##
## Coefficients:
```

##		Estimate	Std. Error	t value	Pr(> t)
##	(Intercept)	1.406768	0.060087	23.41	<2e-16 ***
##	BattingDataSlim\$HR	0.188619	0.003573	52.78	<2e-16 ***

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.81 on 10715 degrees of freedom
## Multiple R-squared:  0.2064, Adjusted R-squared:  0.2063
## F-statistic: 2786 on 1 and 10715 DF,  p-value: < 2.2e-16
```

```
summary(model2)
```

```
##
## Call:
## lm(formula = BattingDataSlim$IBB ~ sqrt(BattingDataSlim$S0))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.830  -2.307  -1.118   0.861  116.931
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      0.17760     0.16019   1.109   0.268
## sqrt(BattingDataSlim$S0)  0.45163     0.01874  24.098 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.165 on 10715 degrees of freedom
## Multiple R-squared:  0.05141,    Adjusted R-squared:  0.05132
## F-statistic: 580.7 on 1 and 10715 DF,  p-value: < 2.2e-16
```

```
summary(model3)
```

```
##
## Call:
## lm(formula = BattingDataSlim$IBB ~ exp(BattingDataSlim$SLG))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.494  -2.209  -0.806   1.123  105.869
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -17.9528     0.4851  -37.01 <2e-16 ***
## exp(BattingDataSlim$SLG)  14.2397     0.3150   45.21 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.919 on 10715 degrees of freedom
## Multiple R-squared:  0.1602, Adjusted R-squared:  0.1601
## F-statistic: 2044 on 1 and 10715 DF,  p-value: < 2.2e-16
```

```
summary(model4)
```

```
##
## Call:
```

```
## lm(formula = BattingDataSlim$IBB ~ (BattingDataSlim$RBI))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.075  -1.964  -0.559   0.999  112.735
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.171088   0.075332   2.271   0.0232 *
## BattingDataSlim$RBI 0.070239   0.001239  56.670   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.751 on 10715 degrees of freedom
## Multiple R-squared:  0.2306, Adjusted R-squared:  0.2305
## F-statistic: 3212 on 1 and 10715 DF, p-value: < 2.2e-16
```

```
confint(model1)
```

```
##              2.5 %    97.5 %
## (Intercept)    1.2889860 1.5245505
## BattingDataSlim$HR 0.1816139 0.1956233
```

```
confint(model2)
```

```
##              2.5 %    97.5 %
## (Intercept)    -0.1363972 0.4915882
## sqrt(BattingDataSlim$S0) 0.4148897 0.4883631
```

```
confint(model3)
```

```
##              2.5 %    97.5 %
## (Intercept)    -18.90374 -17.00179
## exp(BattingDataSlim$SLG) 13.62228 14.85705
```

```
confint(model4)
```

```
##              2.5 %    97.5 %
## (Intercept)    0.02342309 0.31875390
## BattingDataSlim$RBI 0.06780938 0.07266839
```

Bootstrapping (Question 7)

```
model4boot <- Boot(model4, R=1000)
```

```
## Loading required namespace: boot
```

```
summary(model4boot)
```

```
##
## Number of bootstrap replications R = 1000
##              original    bootBias    bootSE    bootMed
## (Intercept)      0.171088  1.8742e-04 0.0790003 0.171693
## BattingDataSlim$RBI 0.070239 -1.3499e-05 0.0018784 0.070219
```

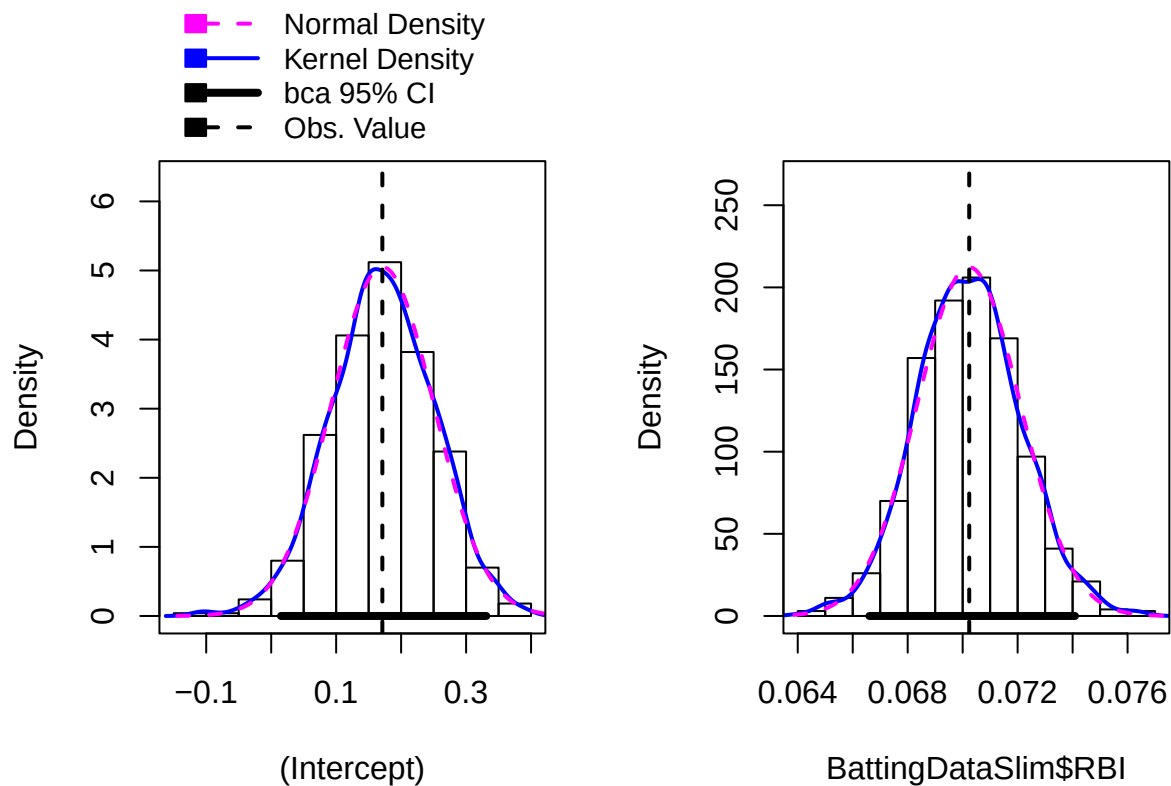
```
confint(model4boot)
```

```
## Warning in confint.boot(model4boot): BCa method fails for this problem. Using
## 'perc' instead
```

```
## Bootstrap percent confidence intervals
##
##              2.5 %      97.5 %
## (Intercept)      0.01501528 0.33118070
## BattingDataSlim$RBI 0.06660627 0.07408782
```

```
hist(model4boot)
```

```
## Warning in confint.boot(x, type = ci, level = level): BCa method fails for this
## problem. Using 'perc' instead
```



```

regression_model <- parsnip::linear_reg() %>%
  parsnip::set_engine("lm") %>%
  parsnip::set_mode("regression")

fit <- regression_model %>%
  parsnip::fit(data = BattingDataSlim, IBB ~ RBI)

summary(fit$fit)

```

```

##
## Call:
## stats::lm(formula = IBB ~ RBI, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.075  -1.964  -0.559   0.999  112.735
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.171088   0.075332   2.271  0.0232 *
## RBI          0.070239   0.001239  56.670 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.751 on 10715 degrees of freedom
## Multiple R-squared:  0.2306, Adjusted R-squared:  0.2305
## F-statistic: 3212 on 1 and 10715 DF, p-value: < 2.2e-16

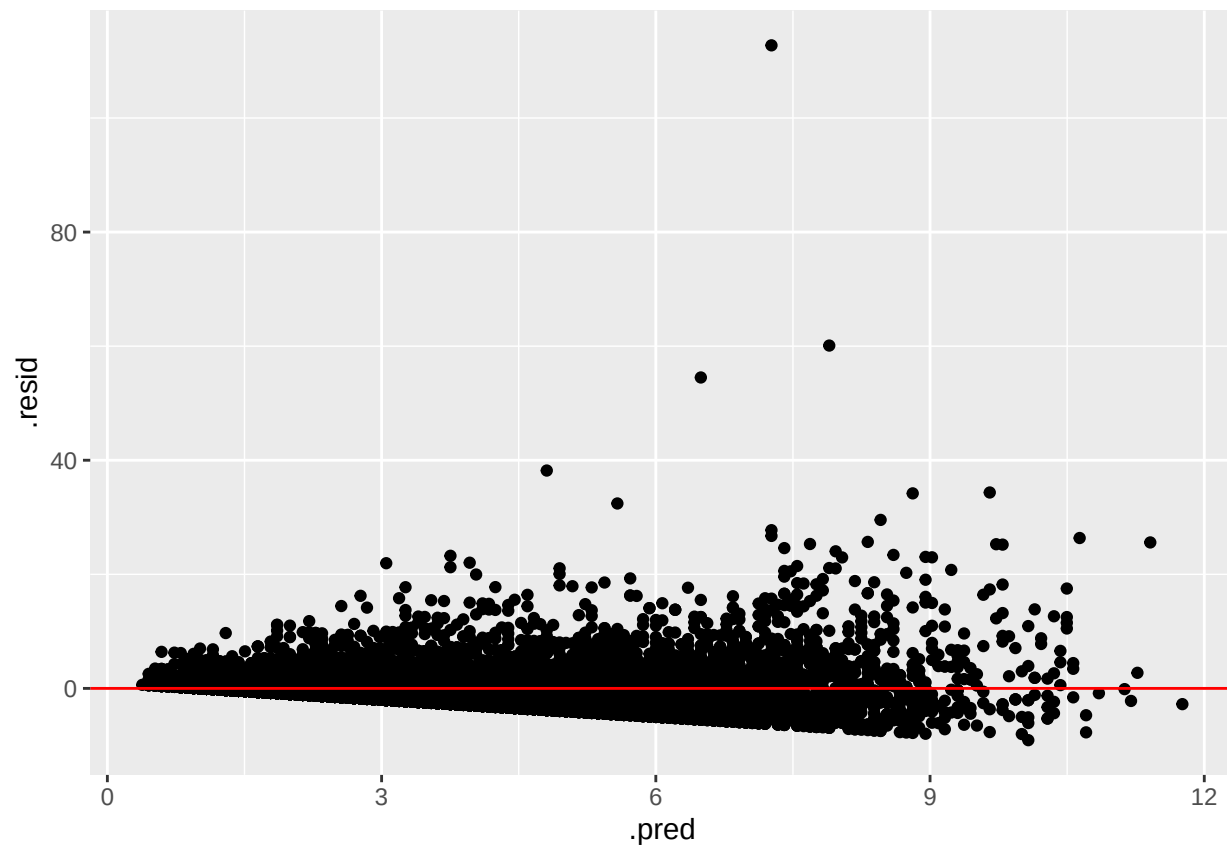
```

```

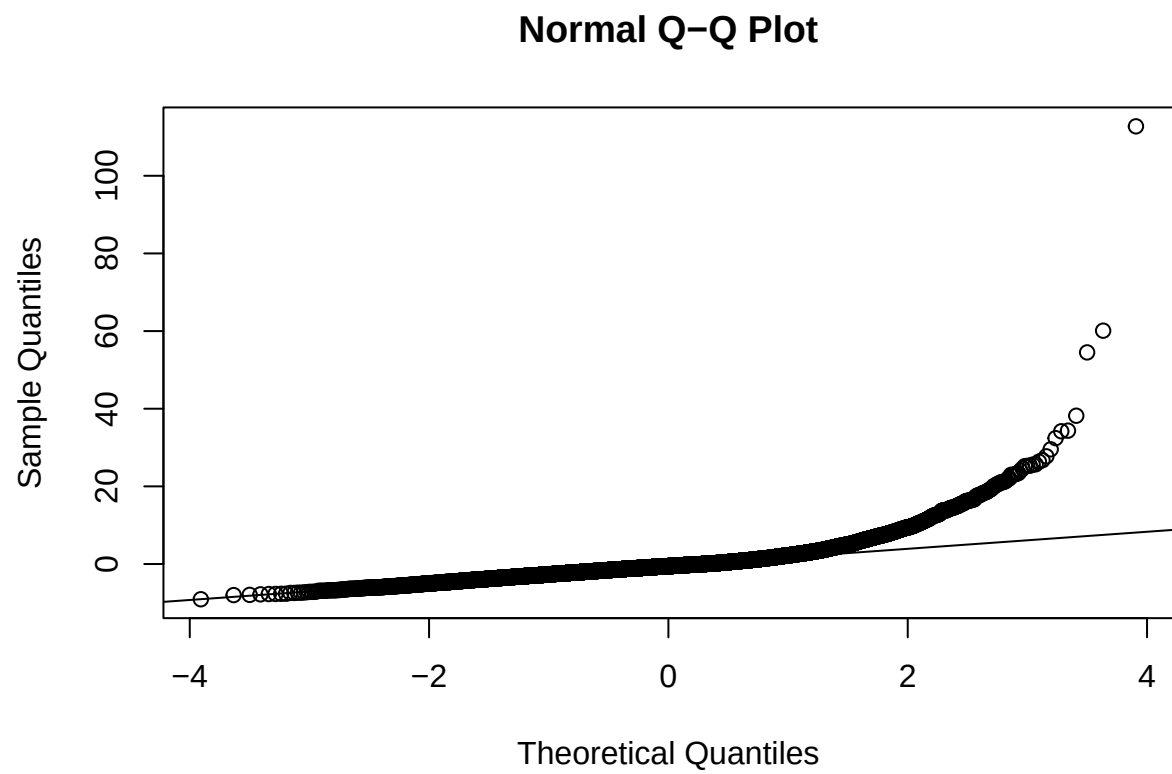
resid <- broom::augment(fit, BattingDataSlim)

ggplot(data = resid, aes(x = .pred, y = .resid)) +
  geom_point() +
  geom_hline(yintercept = 0, color = "red")

```



```
qqnorm(residuals(model4))  
qqline(residuals(model4), lwd = 1)
```

Analysis of Code

The following section answers each of the questions explicitly, while also providing commentary on the important graphs